

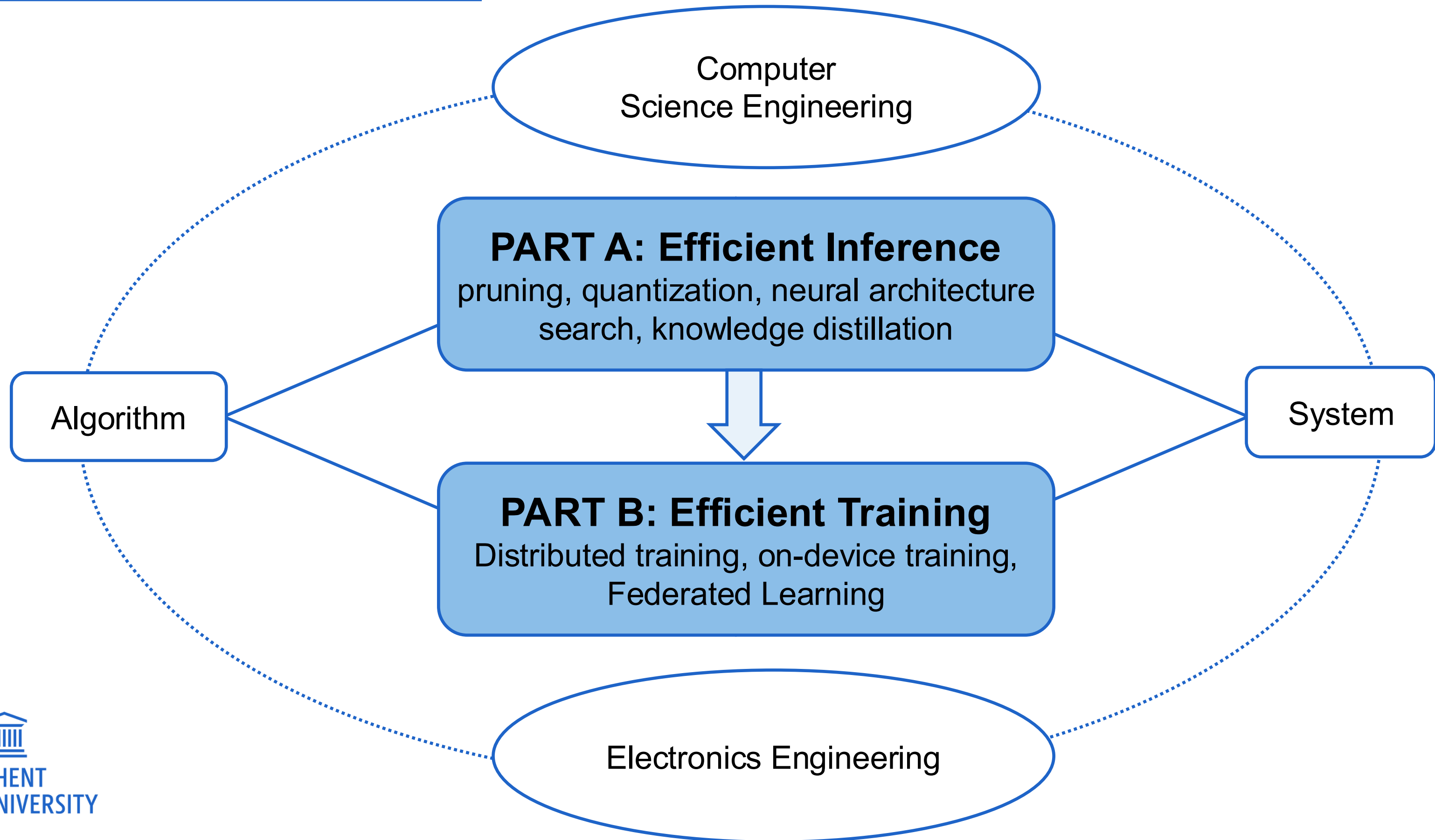


**GHENT
UNIVERSITY**

Embedded machine learning (E061380)

Lecture – 08

Course overview



Lecture plan

Lecture	Topic
Lecture 1 – 13th Feb	Introduction
Lecture 2 – 20th Feb	Overview of Embedded Systems and ML/DL Overflow
Lecture 3 – 27th Feb	Pruning and Sparsity
Lecture 4 – 6th Mar	Quantization
Lecture 5 – 13th Mar	Lab Pruning and Quantization
Lecture 6 – 27th Mar	Neural Architecture Search
Lecture 7 – 3rd April	Lab Pruning and Quantization on Nano 33 BLE
Lecture 8 – 24th April	Knowledge Distillation
Lecture 9 – 8th May	Distributed training, on-device learning and Transfer Learning
Lecture 10 – 13th May	Lab Knowledge Distillation and Federated Learning

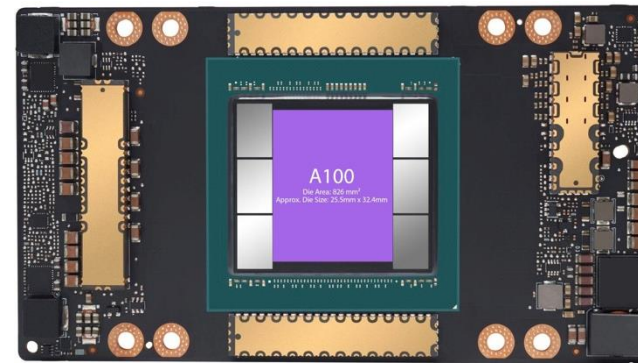
Recap of Last Lecture

- Primitive operations: Linear layer, convolution, grouped convolution, depth-wise convolution
- Bottleneck architecture, ShuffleNet
- Neural architecture search
- Network-level search space: depth, resolution, width, kernel size, topology connection
- Search strategy: grid search, random search, reinforcement learning, gradient descent, evolutionary search
- General models to specialized models

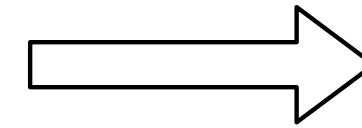
Lecture Outline

- Remaining part of Neural Architecture Search
- What is knowledge distillation
- What to match
 - Output logits
 - Intermediate weights
 - Intermediate features
 - Gradients
 - Sparsity patterns
 - Relational information
- Self and online distillation
 - Self Distillation
 - Online Distillation
 - Combined
- Network augmentation

Challenge: Limited Hardware Resources



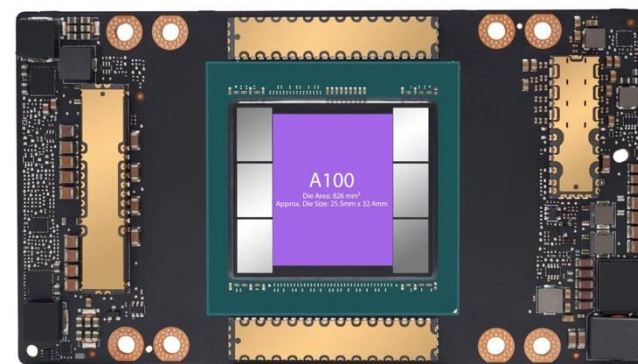
Cloud AI



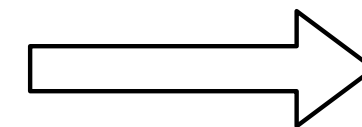
Tiny AI

Computation (fp32)	19.5 TFLOPS	MFLOPs
Memory	80GB	256kB

Challenge: Limited Hardware Resources



Cloud AI



Tiny AI

Computation (fp32)	19.5 TFLOPS	MFLOPs
Memory	80GB	256kB
Neural Network	ResNet ViT-Large	MCUNet MobileNetV2-Tiny

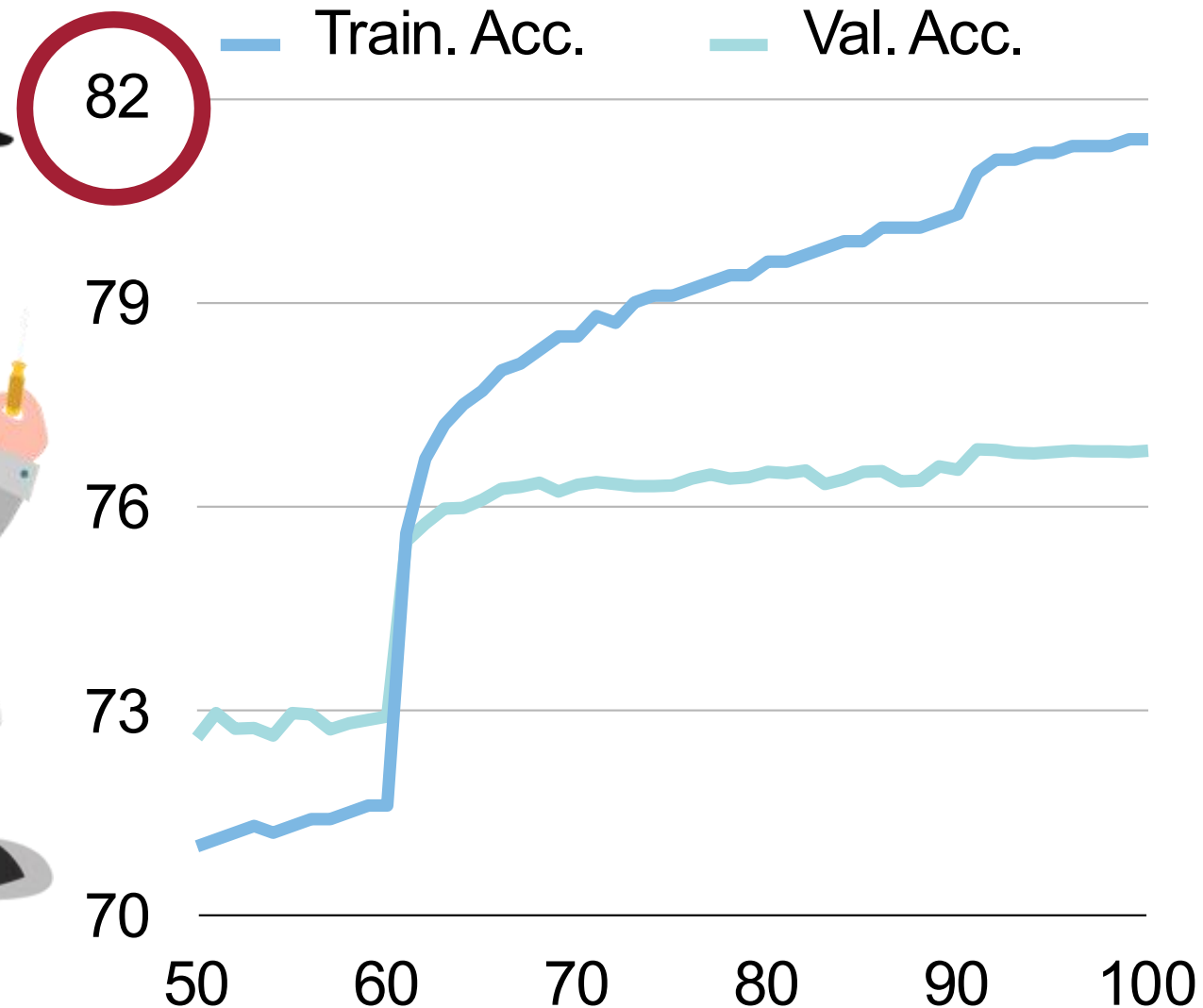
- Neural network must be **tiny** to run efficiently on tiny edge devices. How to train tiny model with the help of large model?

Tiny Models are Hard to Train

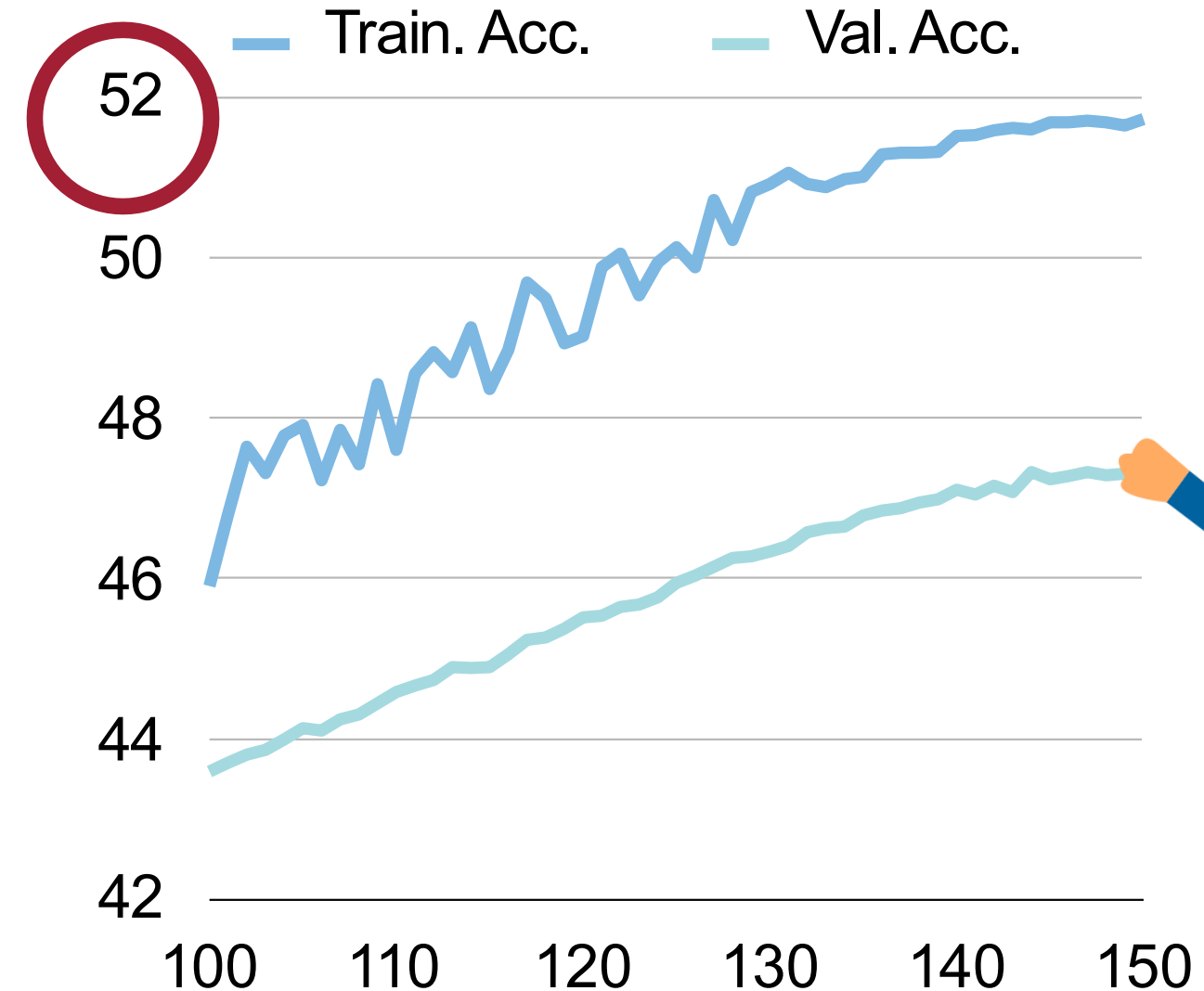
vaak underfitten

Tiny models underfit large datasets

Training curve for ResNet50

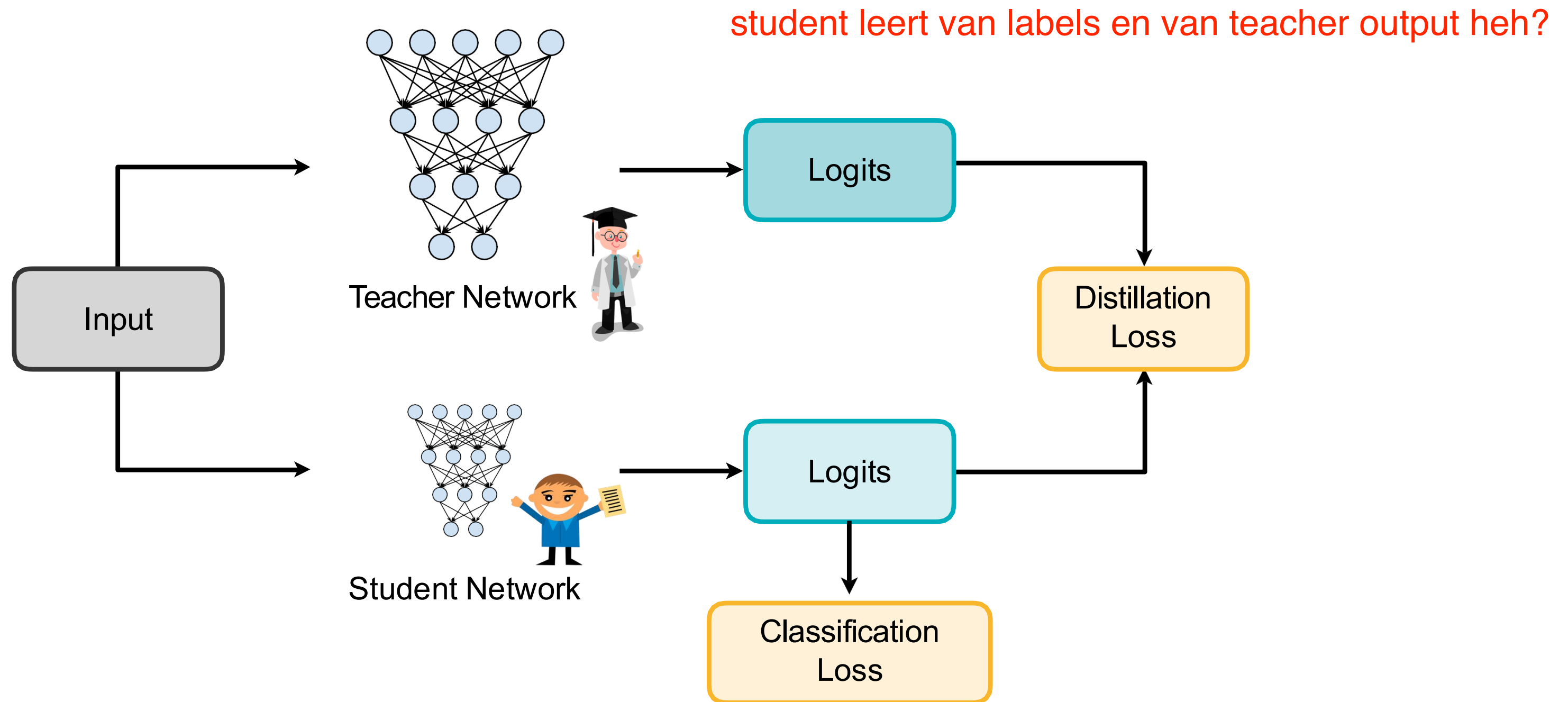


Training curve for MobileNetV2-Tiny



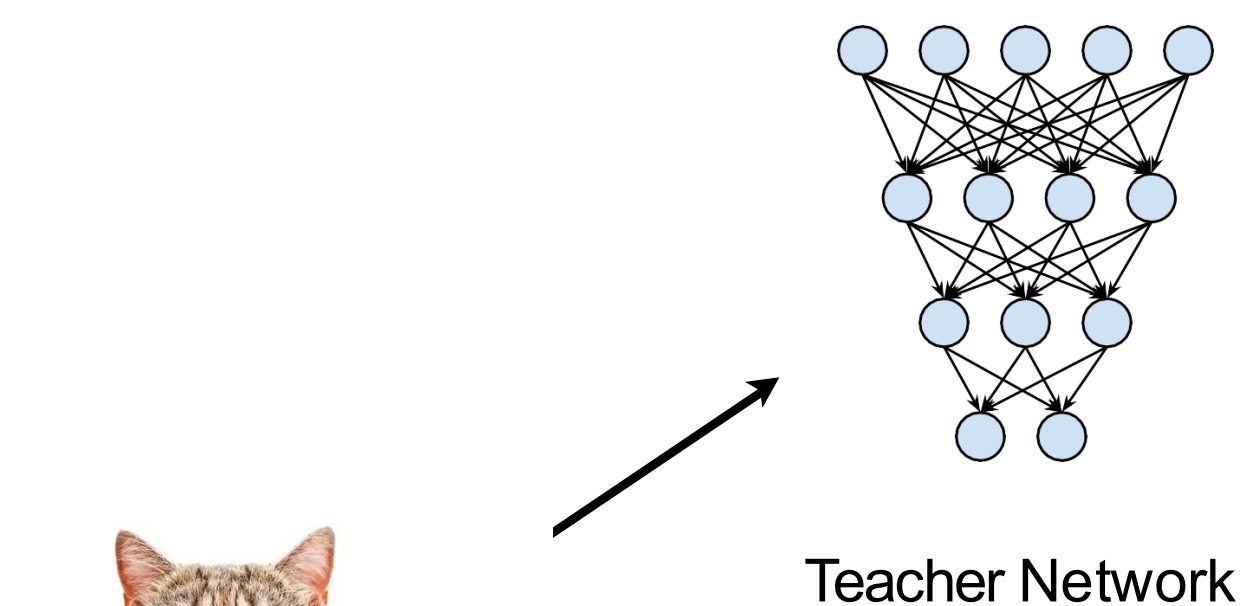
Question: Can we help the training of tiny models with large models?

Illustration of Knowledge Distillation



Intuition of Knowledge Distillation

Matching prediction probabilities between teacher and student

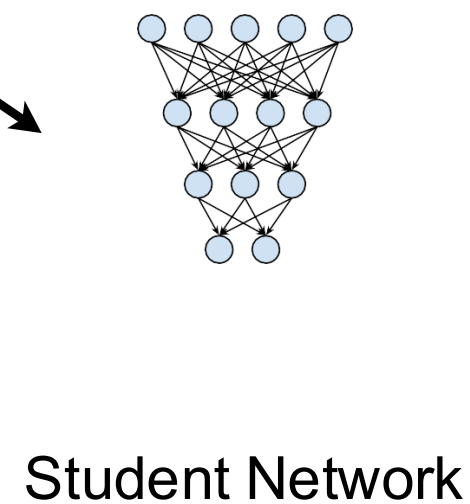


Teacher Network

	Logits	Probabilities
Cat	5	0.982
Dog	1	0.017

**Softmax
Function**

$$\frac{\exp(5)}{\exp(5) + \exp(1)}$$
$$\frac{\exp(1)}{\exp(5) + \exp(1)}$$



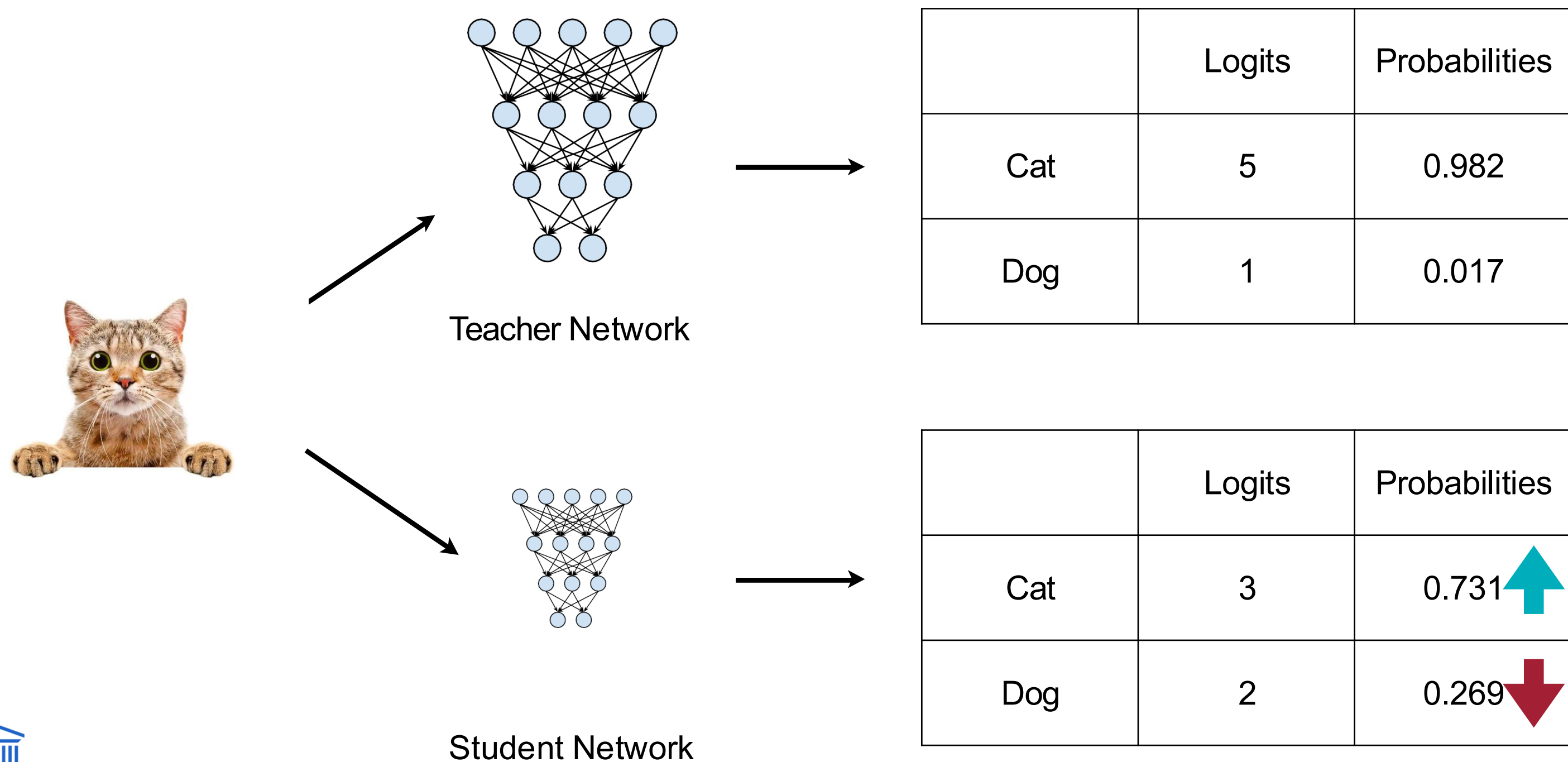
Student Network

	Logits	Probabilities
Cat	3	0.731
Dog	2	0.269

The student model is less confident

Intuition of Knowledge Distillation

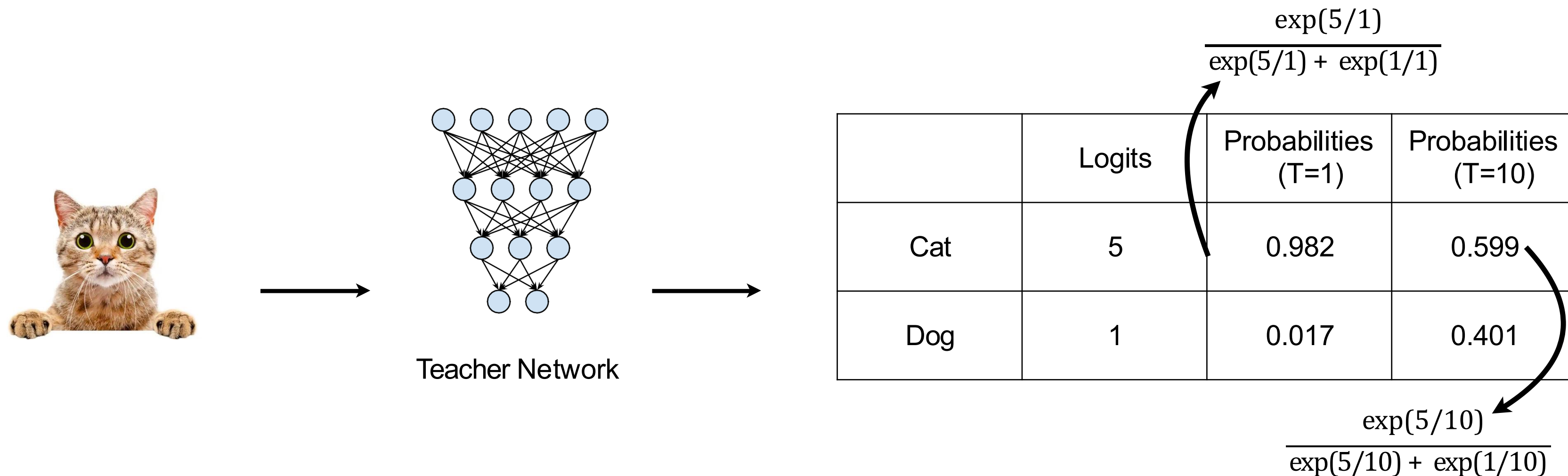
Matching prediction probabilities between teacher and student



soft probabilities bevatten extra informatie over klasse-relaties

Intuition of Knowledge Distillation

Concept of temperature



A larger temperature smooths the output probability distribution.

Formal Definition of KD

- Neural networks typically use a softmax function to generate the **logits** z_i to class **probabilities**
$$p(z_i, T) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$
. Here, $i, j = 0, 1, 2, \dots, C - 1$, where C is the number of classes. T is the temperature, which is normally set to 1.
- The goal of knowledge distillation is to **align the class probability distributions from teacher and student networks**.

aka: een klein model laten leren van een groter, slimmer model
student leer niet alleen van het echte label, maar ook van wat de teacher denkt

bv: foto van een kat

label zegt: kat = 1, dog = 0, car = 0

wat teacher zegt: cat = 0.85, dog = 0.12, car = 0.03 -> beter om relaties te leren: kat lijkt beetje op hond, niet op auto

die extra kennis = dark knowledge

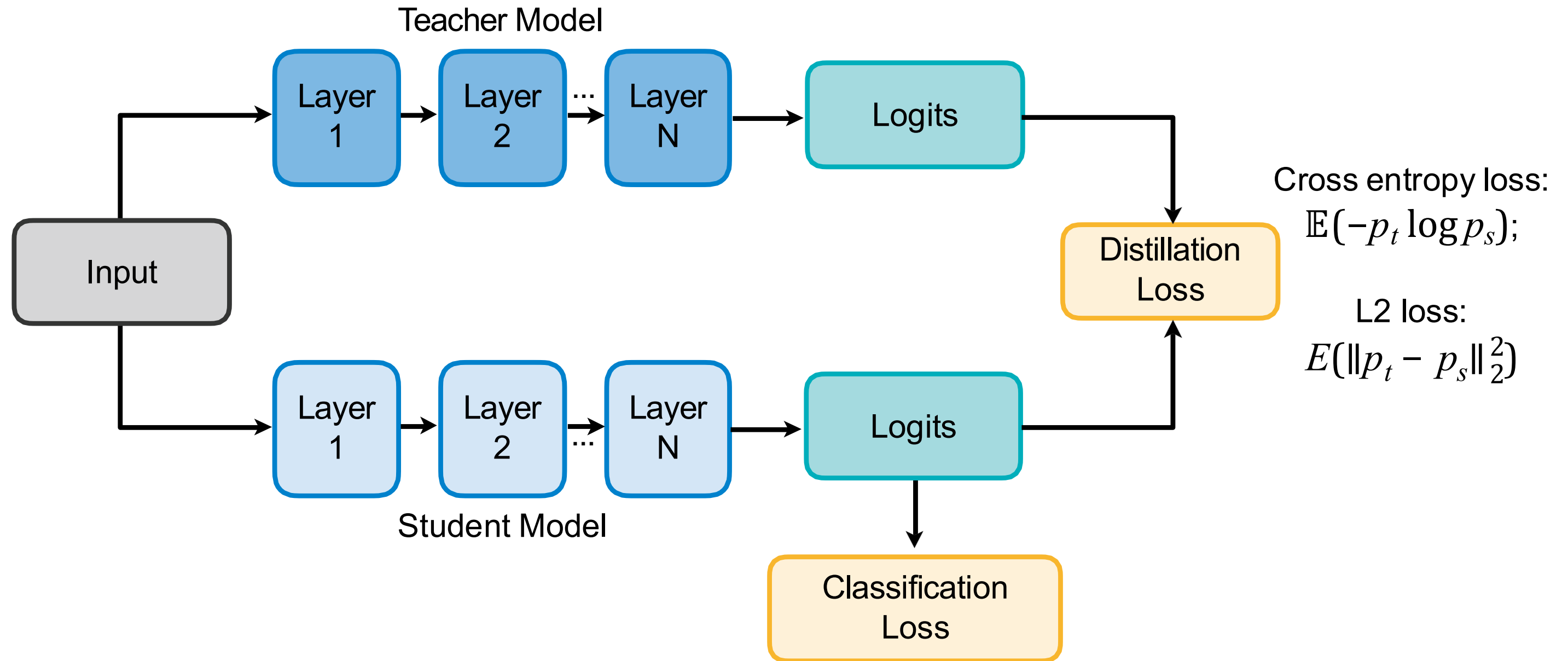
student probeert dan niet alleen kat te voorspellen, maar de volledige verdeling van teacher na te bootsen

Lecture Outline

- What is knowledge distillation
- What to match
 - Output logits
 - Intermediate weights
 - Intermediate features
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 - Sparsity patterns
 - Relational information
- Self and online distillation
 - Self Distillation
 - Online Distillation
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- Network augmentation

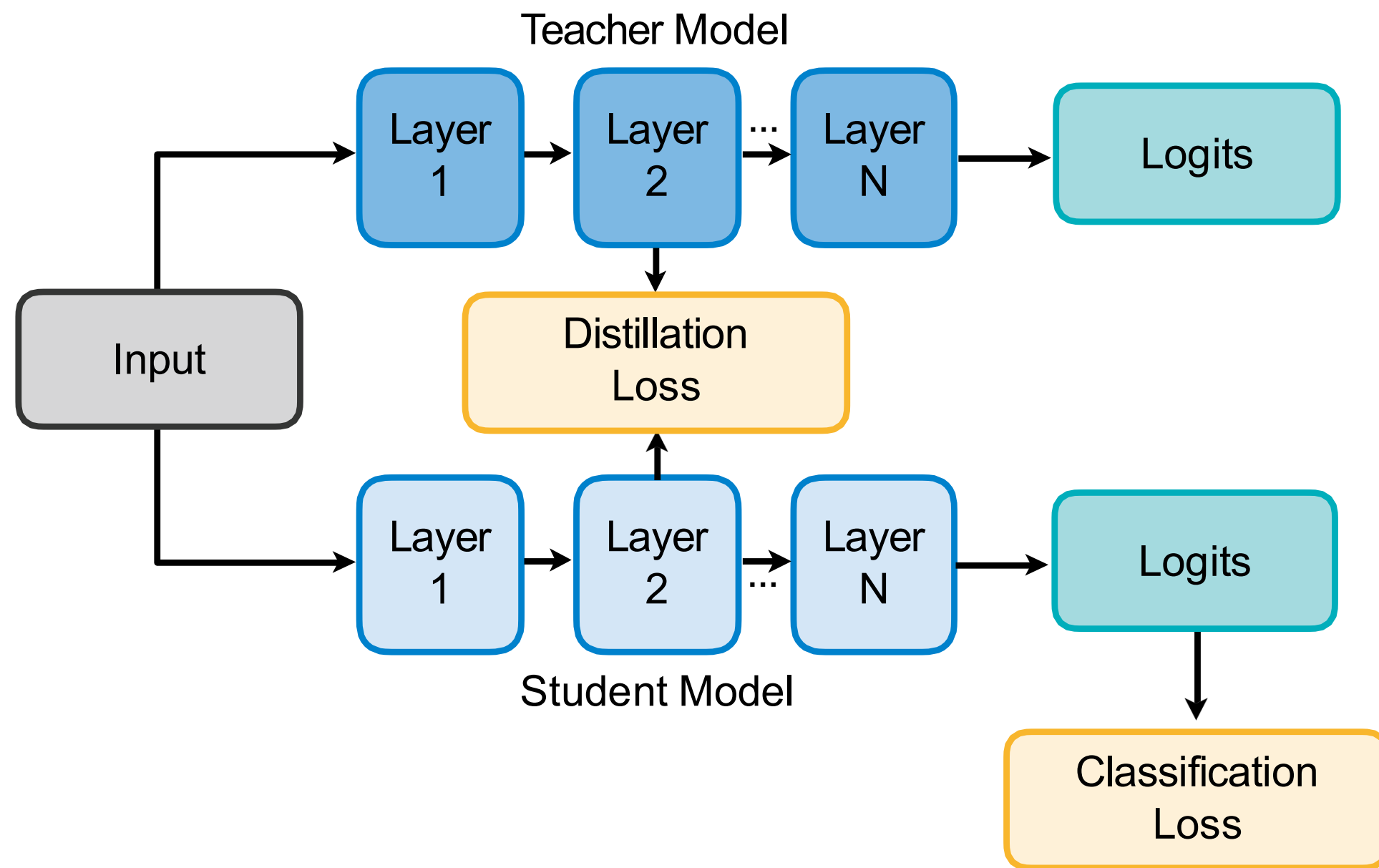
Matching Output Logits

standaard KD matcht output logits / probabilitites
hier loss: bv cross entropy / l2



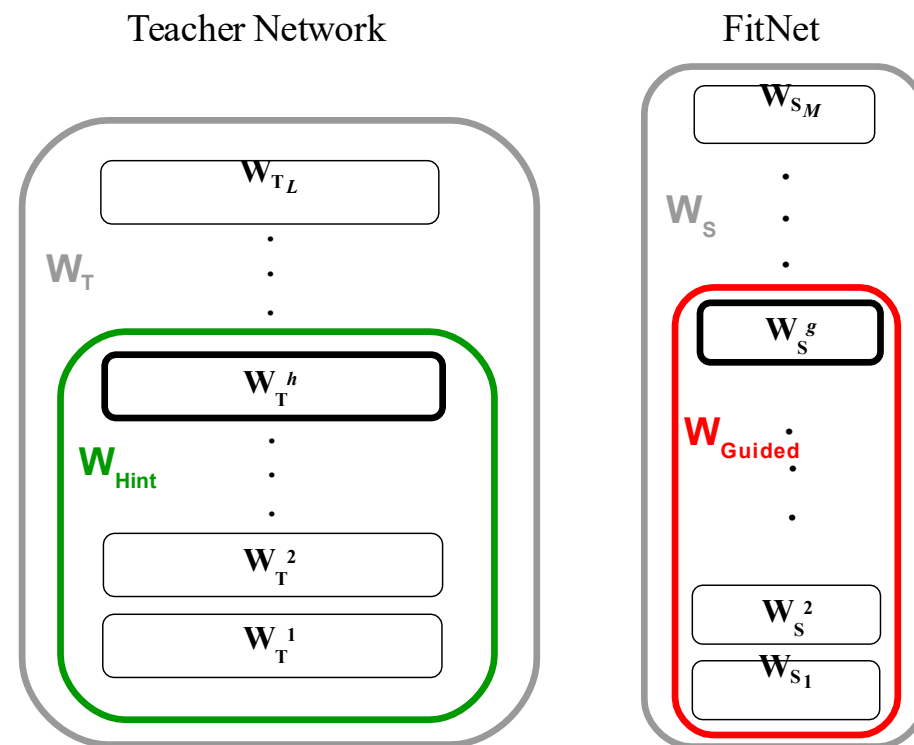
What else to match other than output logits?

Matching intermediate weights

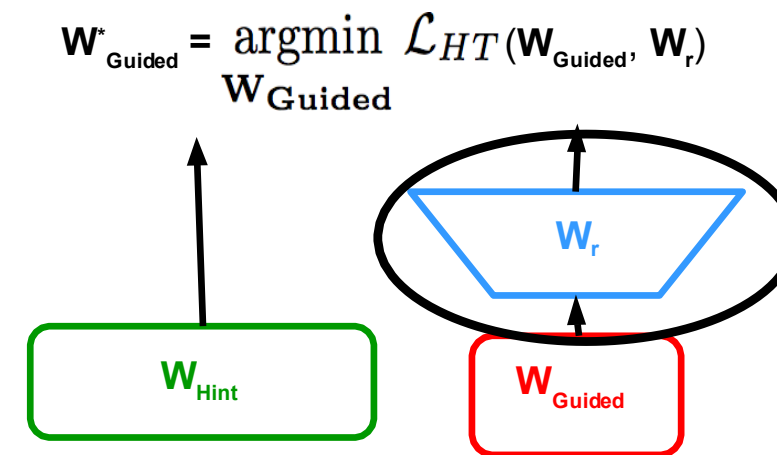


Matching intermediate weights

teacher en student vaak verschillende dimensies: lineaire laag om de shapes te aligneren



(a) Teacher and Student Networks



(b) Hints Training

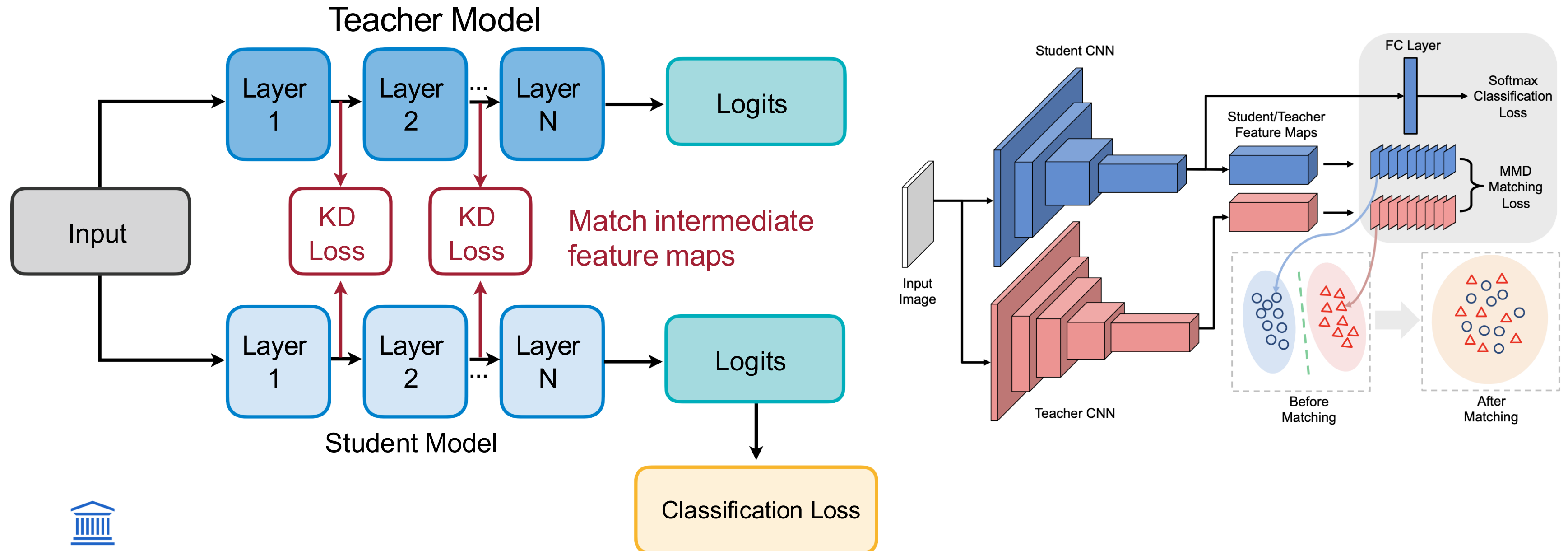
An FC layer used to align the shapes of teacher and student weights

- Other than the cross-entropy distillation loss, also add a L2 loss between teacher weights and student weights (linear transformation is applied to match the dimensionalities).

Matching intermediate features

Minimizing maximum mean discrepancy between feature maps

- Intuition: teacher and student networks should have similar **feature** distributions, not just output probability distributions.



Intermediate attention maps

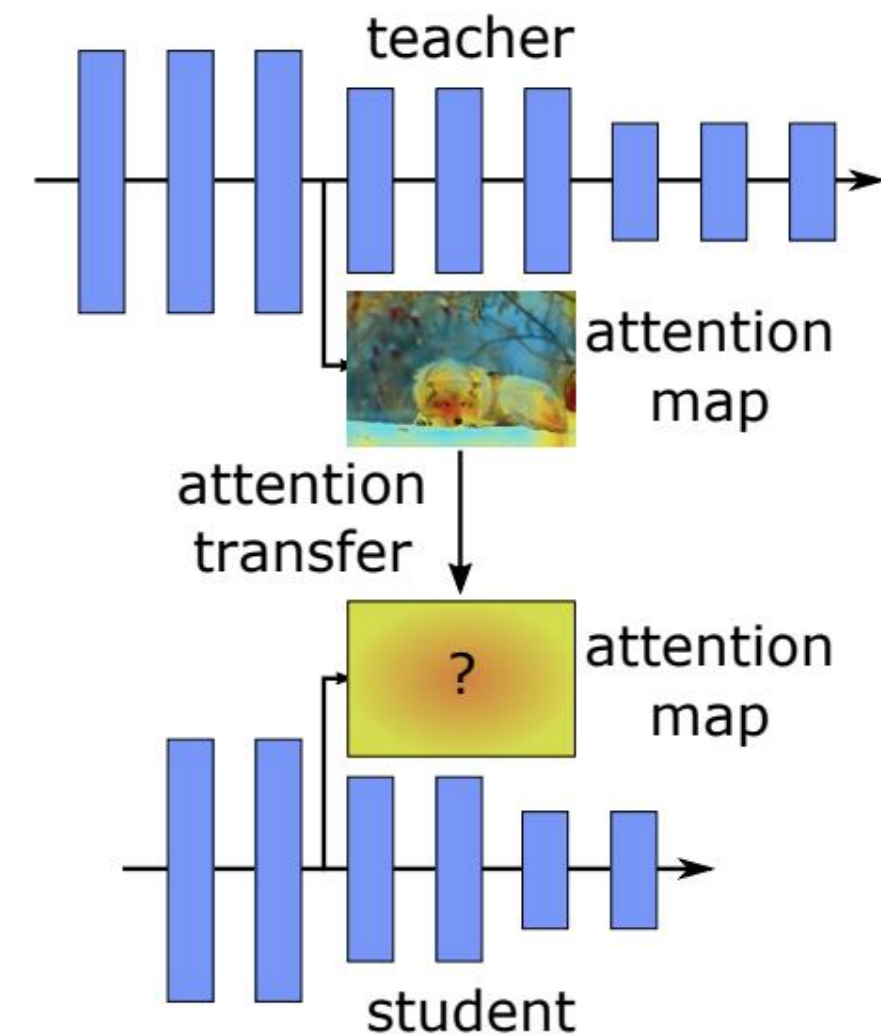
Gradients of feature maps are used to characterize “attention” of DNNs

- The attention of a CNN feature map x is defined as $\frac{\partial L}{\partial x}$, where L is the learning objective.
- Intuition: If $\frac{\partial L}{\partial x_{i,j}}$ is large, a small perturbation at i, j will significantly impact the final output. As a result, the network is putting more attention on position i, j .

input image



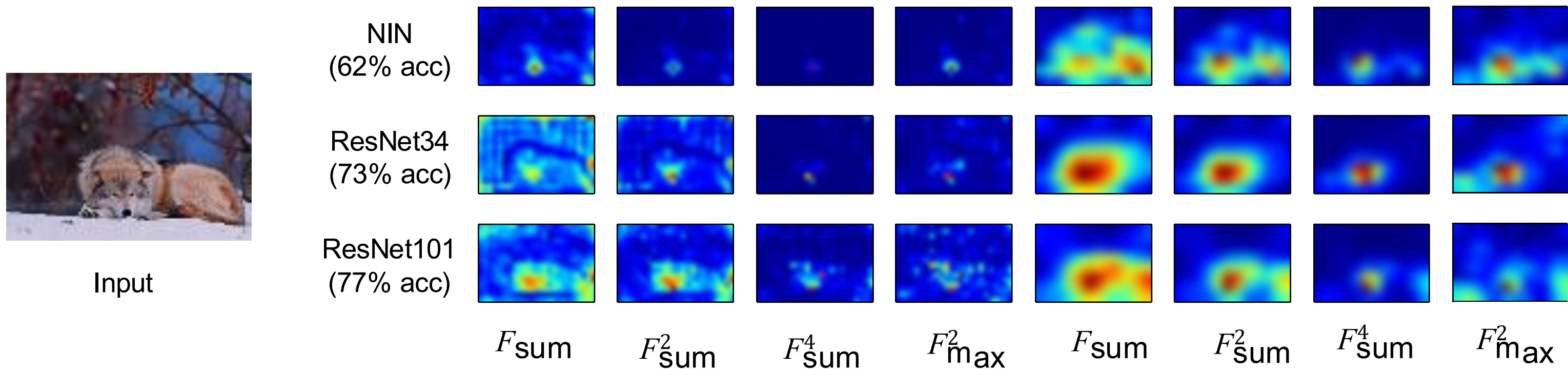
attention map



Intermediate attention maps

Performant models have similar attention maps

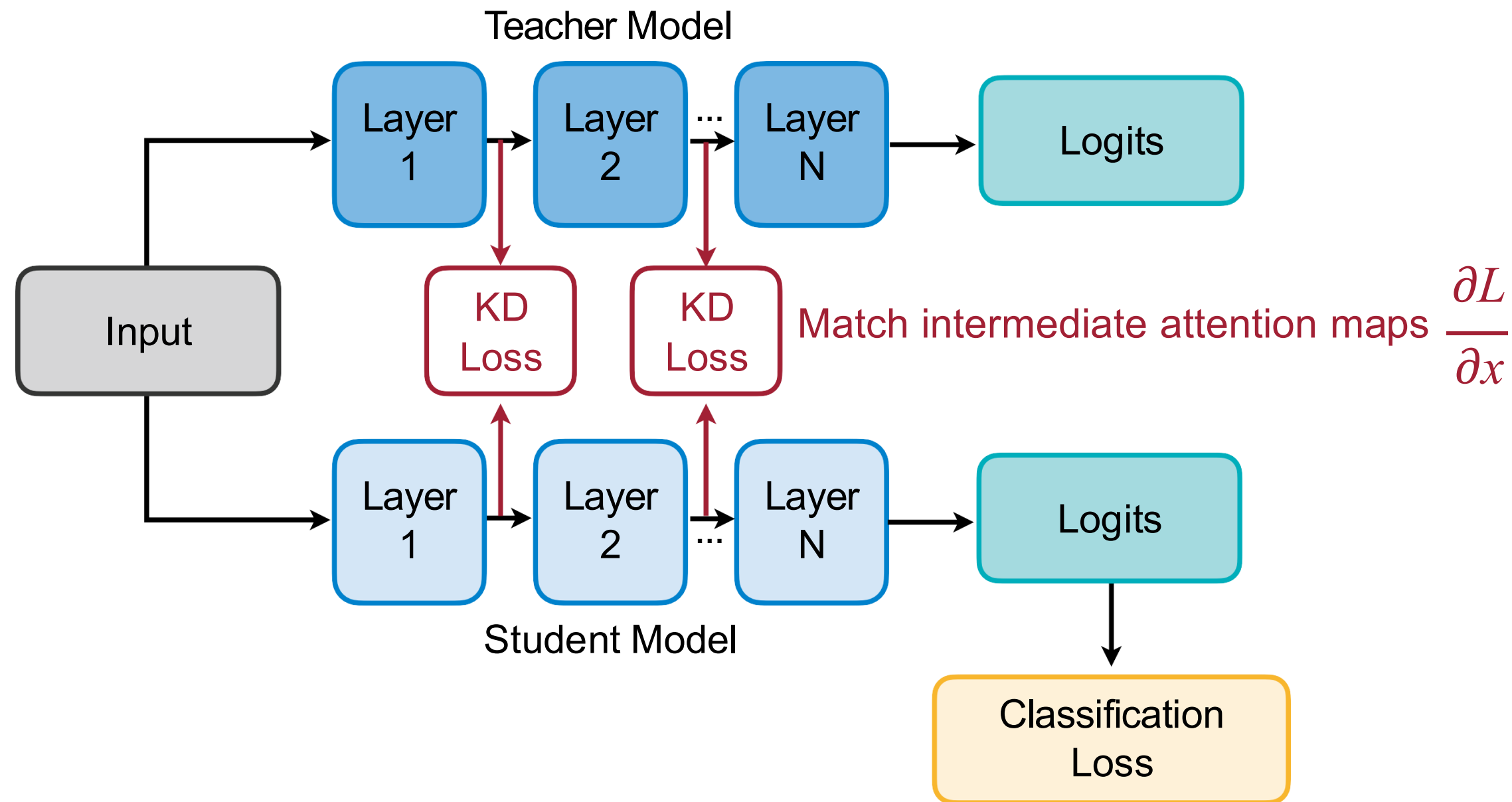
- Attention maps of performant ImageNet models (ResNets) are indeed similar to each other, but the less performant model (NIN) has quite different attention maps.



Different reduction methods across the channel dimensions

Matching intermediate attention maps

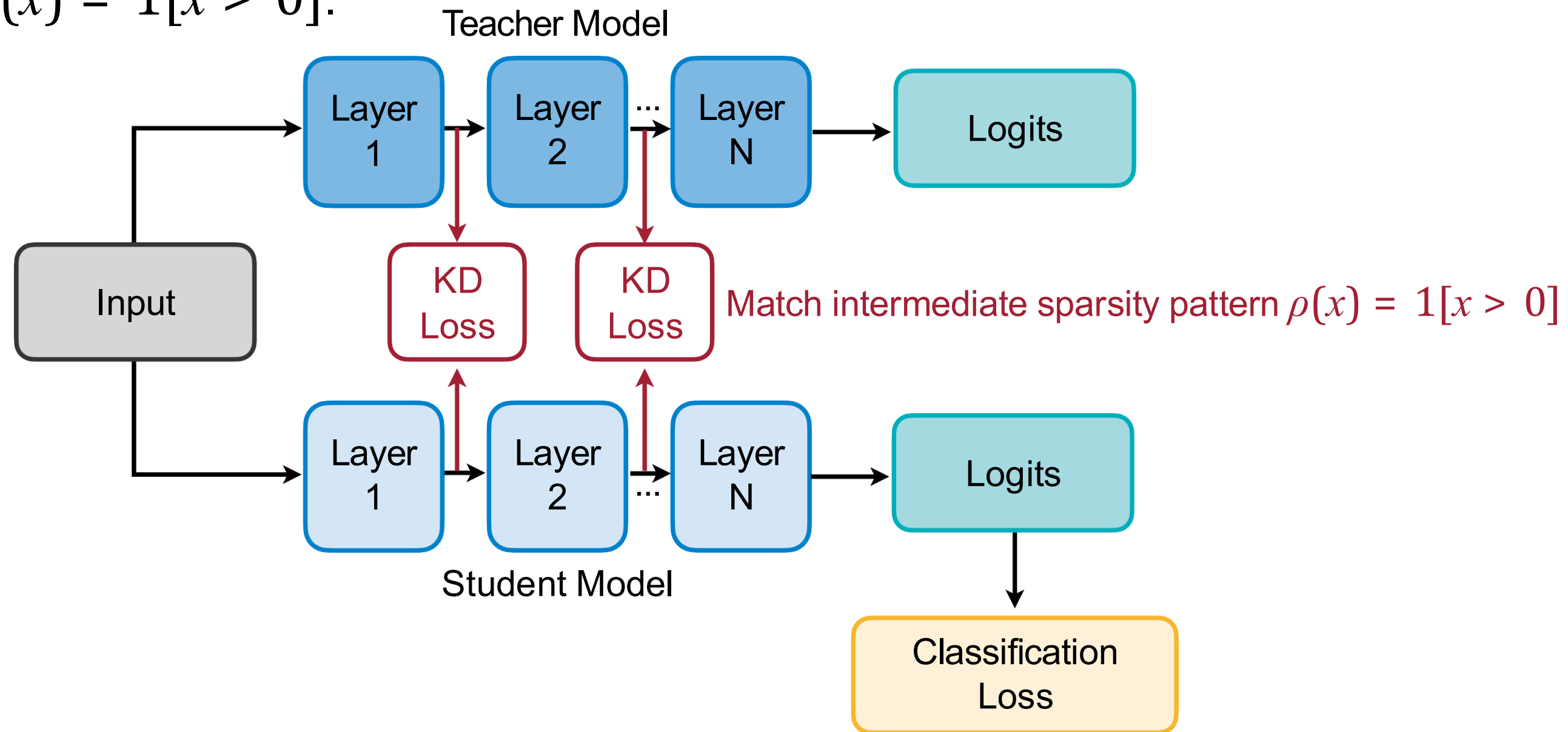
student wordt getraind om attention maps van teacher te matchen
-> kd loss op tussenliggende attention maps



Matching sparsity patterns

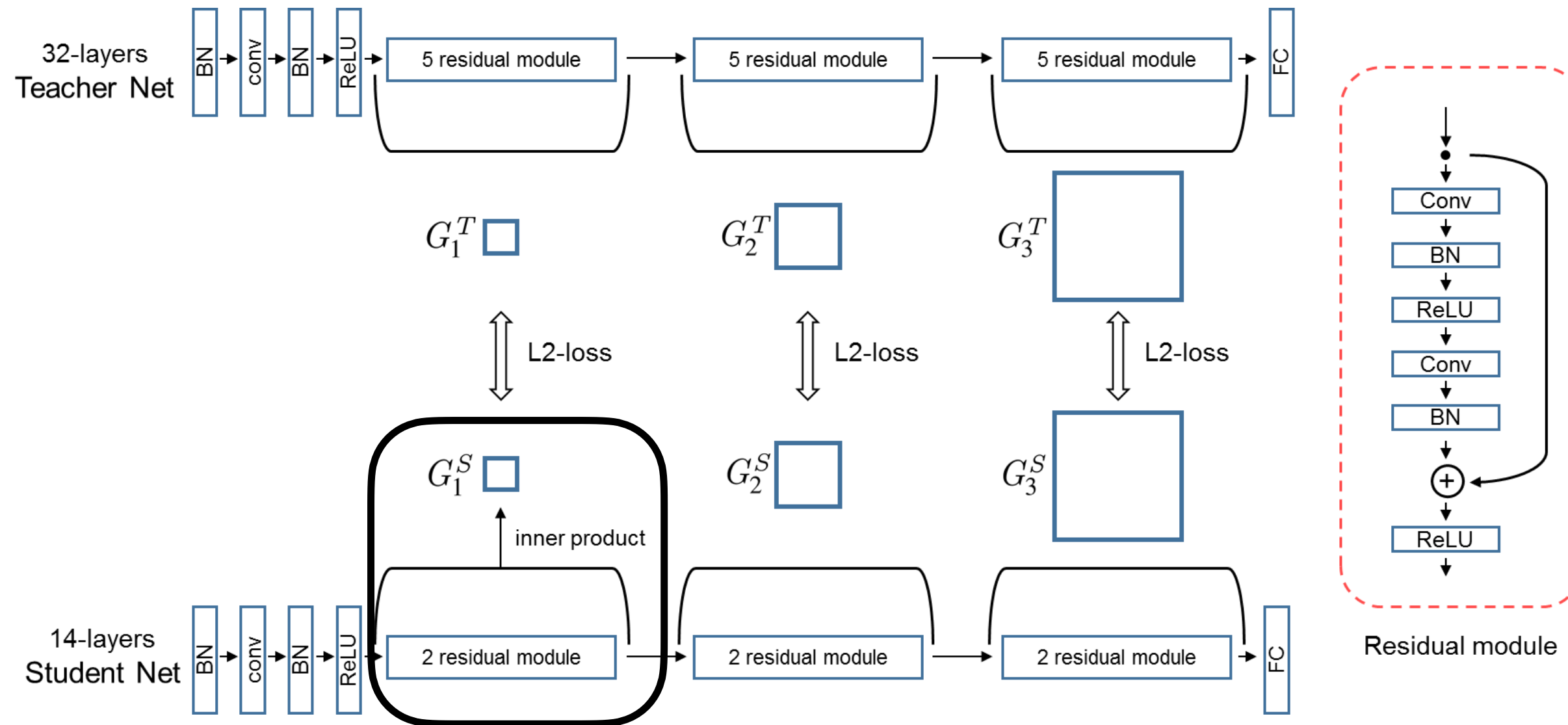
matchen welke neuronen actief zijn na ReLU

- Intuition: the teacher and student networks should have similar sparsity patterns after the ReLU activation. A neuron is **activated** after ReLU if its value is larger than 0, denoted by the indicator function $\rho(x) = 1[x > 0]$.



Matching relational information

Relations between different layers



Use inner product to extract relational information (a matrix of shape $C_{in} \times C_{out}$, reduction on the spatial dimensions) for both student and teacher networks.

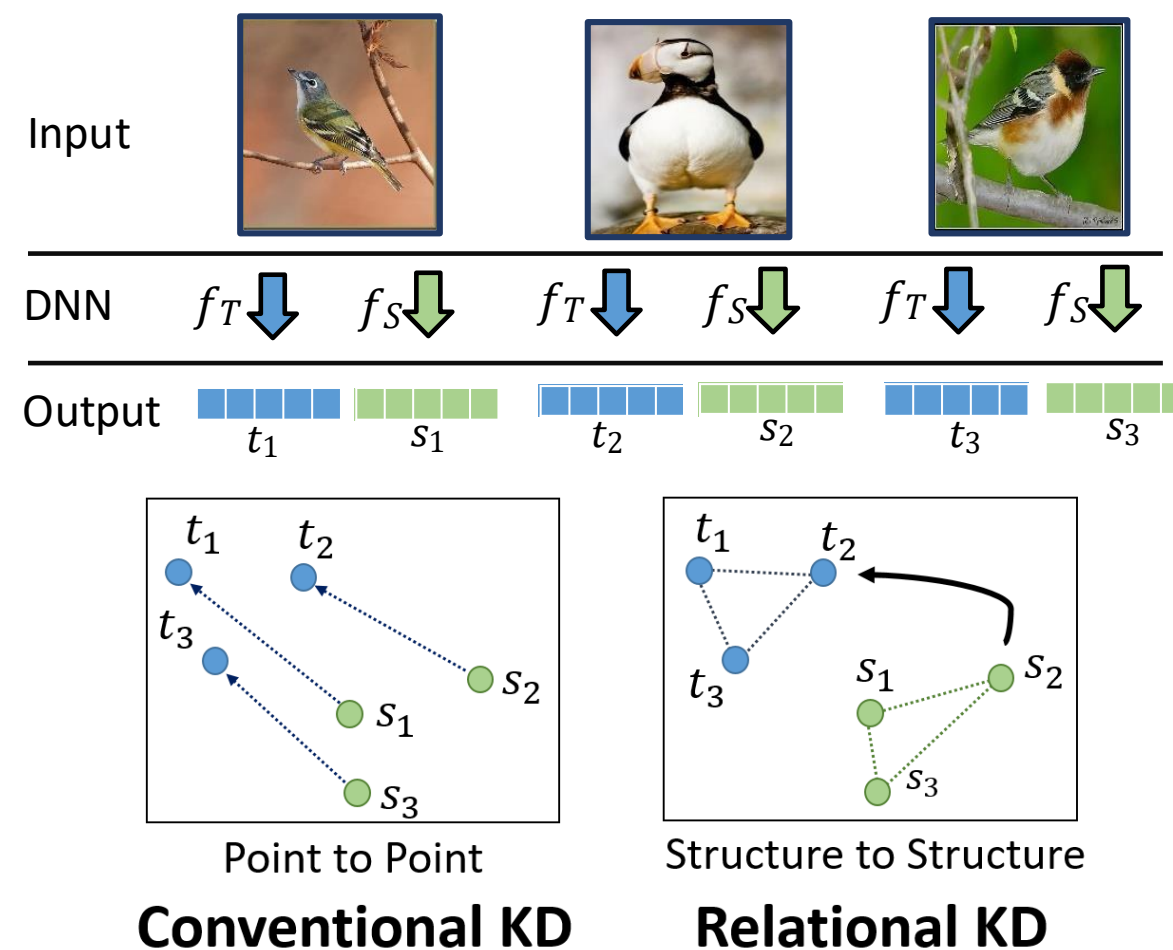
Note: the student and teacher networks only differ in number of layers, not number of channels.

Then match the resulting dot products between teacher and student networks (G_1^T , G_1^S).

A Gift from Knowledge Distillation: Fast Optimization, Network Minimization and Transfer Learning [Yim *et al.*, CVPR 2017]

Matching relational information

Relations between different samples



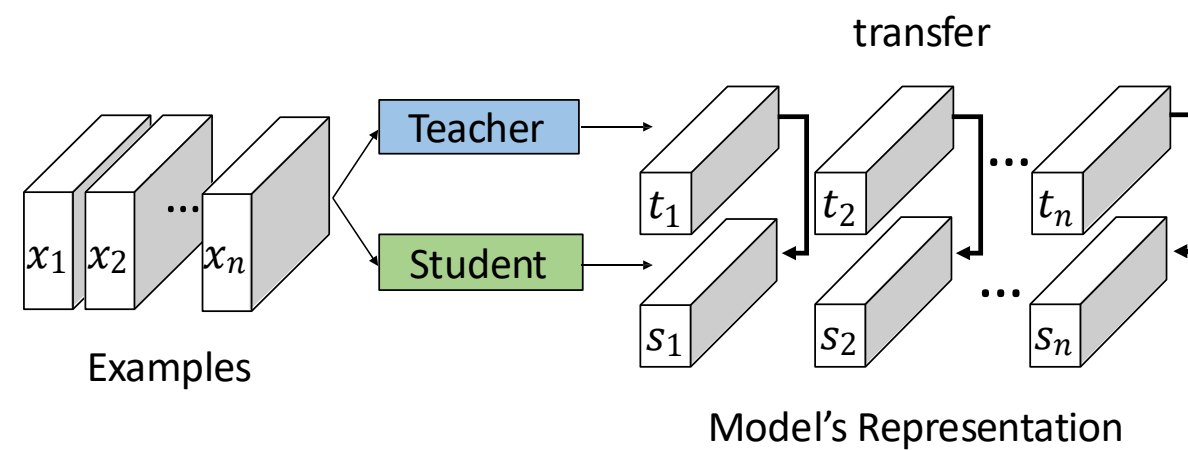
Conventional KD focuses on matching features / logits for **one input**.

Relational KD looks at the **relations** between intermediate features from **multiple inputs**.

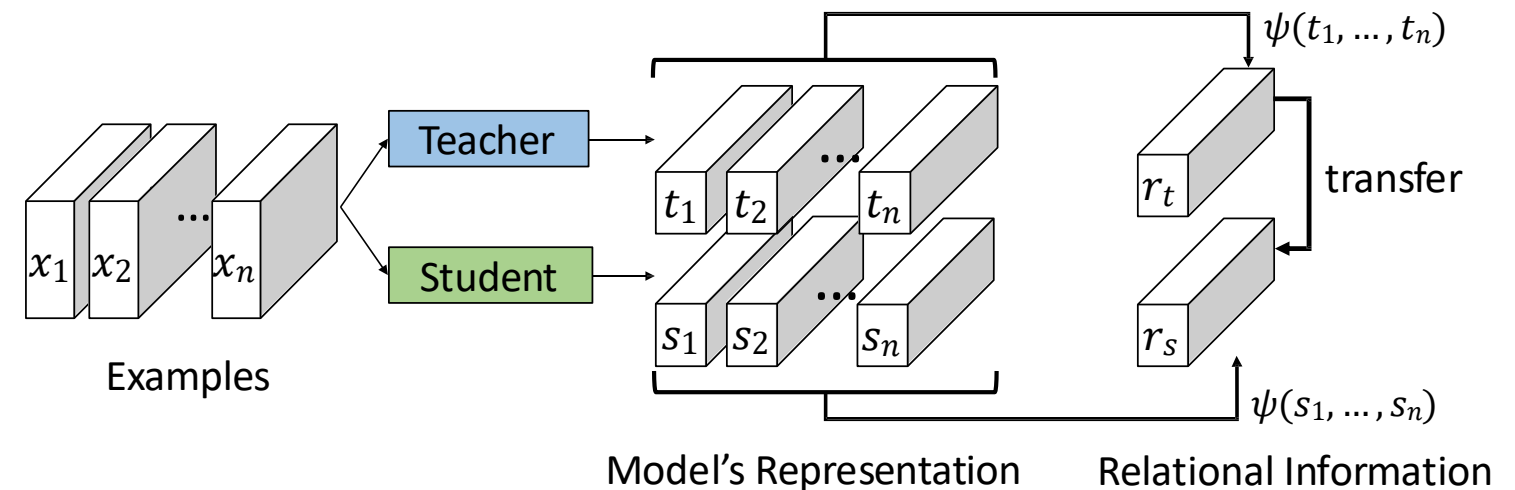
Matching relational information

Relations between different samples

$\psi(s_1, s_2, \dots, s_n) = (\|s_1 - s_2\|_2^2, \|s_1 - s_3\|_2^2, \dots, \|s_1 - s_n\|_2^2, \dots, \|s_{n-1} - s_n\|_2^2)$ is a vector of length $n(n-1)/2$ representing pairwise distances of feature vectors.



Individual Knowledge Distillation



Relational Knowledge Distillation

Relation **within** the model, calculated **separately** for each input

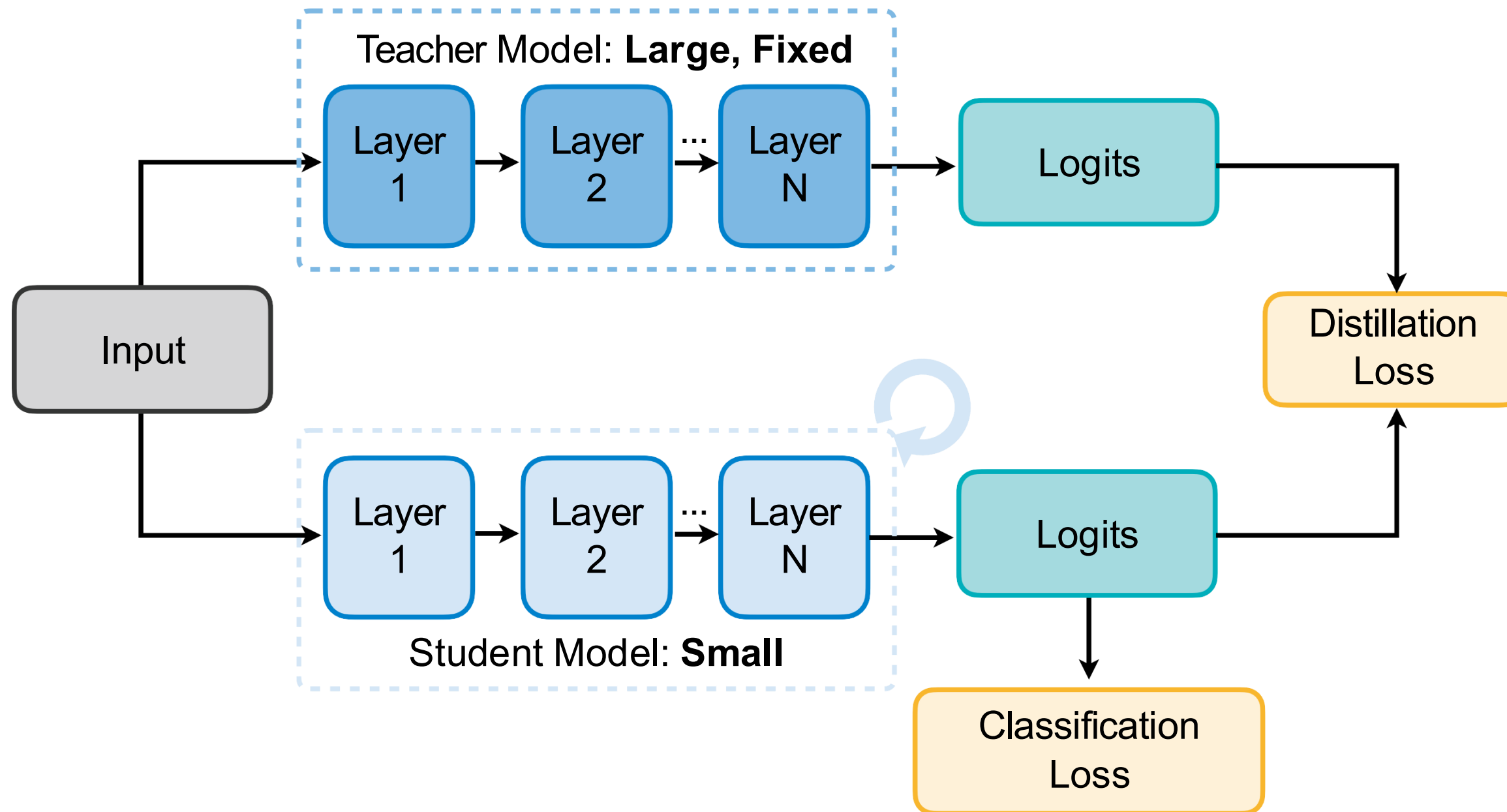
Relation **across different samples**

Lecture Outline

- What is knowledge distillation
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- **Self and online distillation**
 - **Self Distillation**
 - **Online Distillation**
 - **Combined**
- Network augmentation

Overview of knowledge distillation

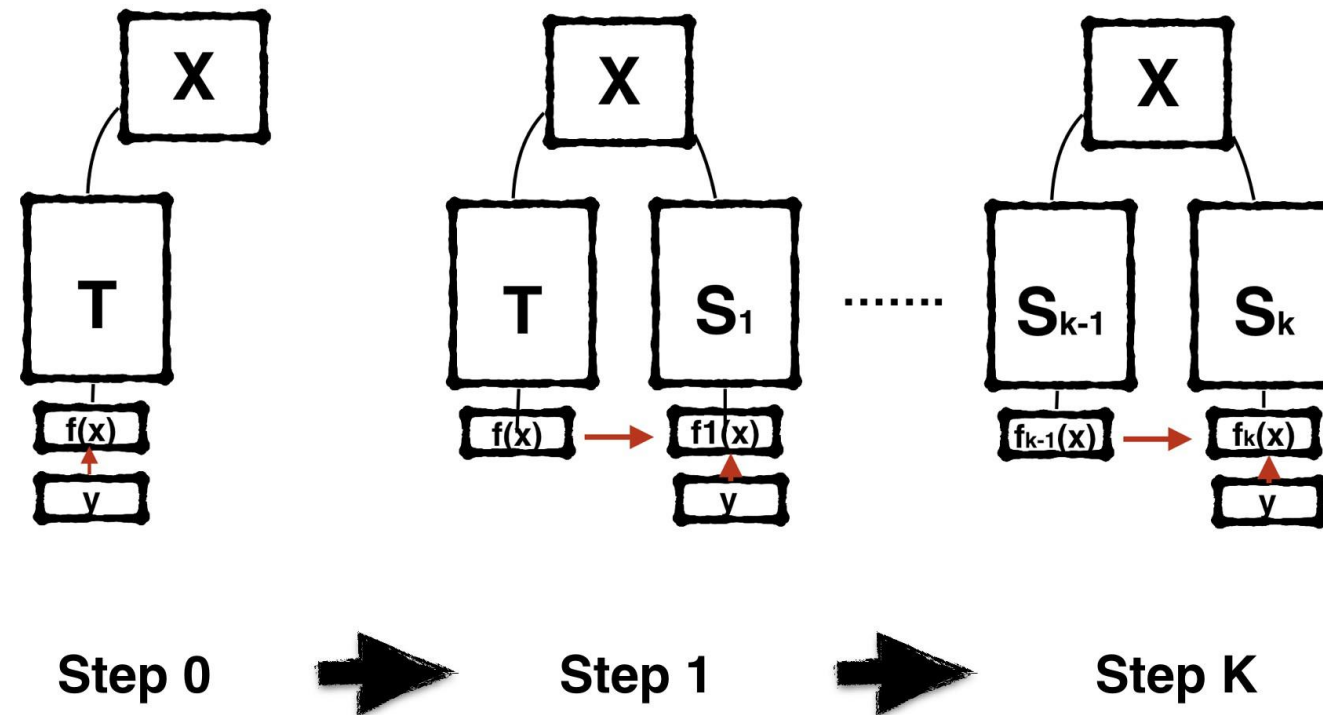
Teacher model is usually larger than the student model and is fixed



Discussion: What is the disadvantage of fixed large teachers? Does it have to be the case that we need a fixed large teacher in KD?

Knowledge Distillation: A Survey [Gou *et al.*, IJCV 2020]

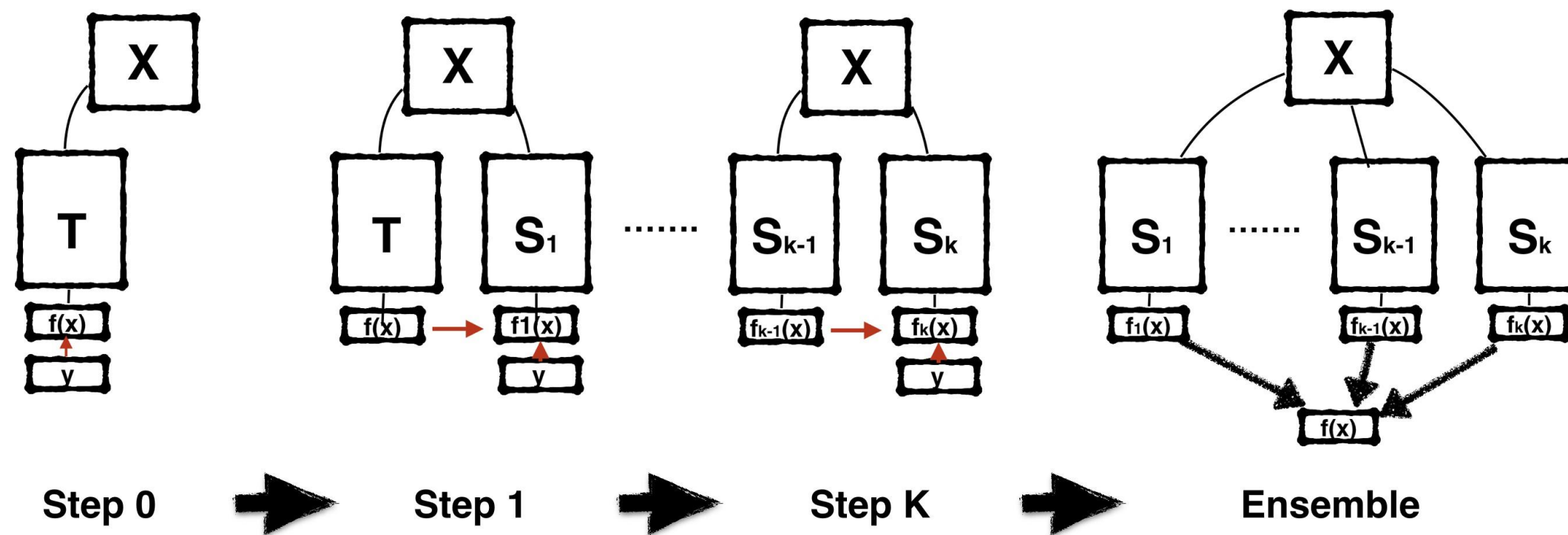
Self-Distillation with Born-Again NNs



hetzelfde netwerk wordt iteratief opnieuw getraind
naast betere individuele studenten kan men ook meerdere generaties ensembles

- Born-Again Networks **adds iterative training stages** and **using both classification objective and distillation objective** in subsequent stages.
- Network architecture $T = S_1 = S_2 = \dots = S_k$.
- Network accuracy $T < S_1 < S_2 < \dots < S_k$.
- Can alternatively ensemble $T, S_1, S_2, \dots, S_k$ to get even better performance.

Self-Distillation with Born-Again NNs



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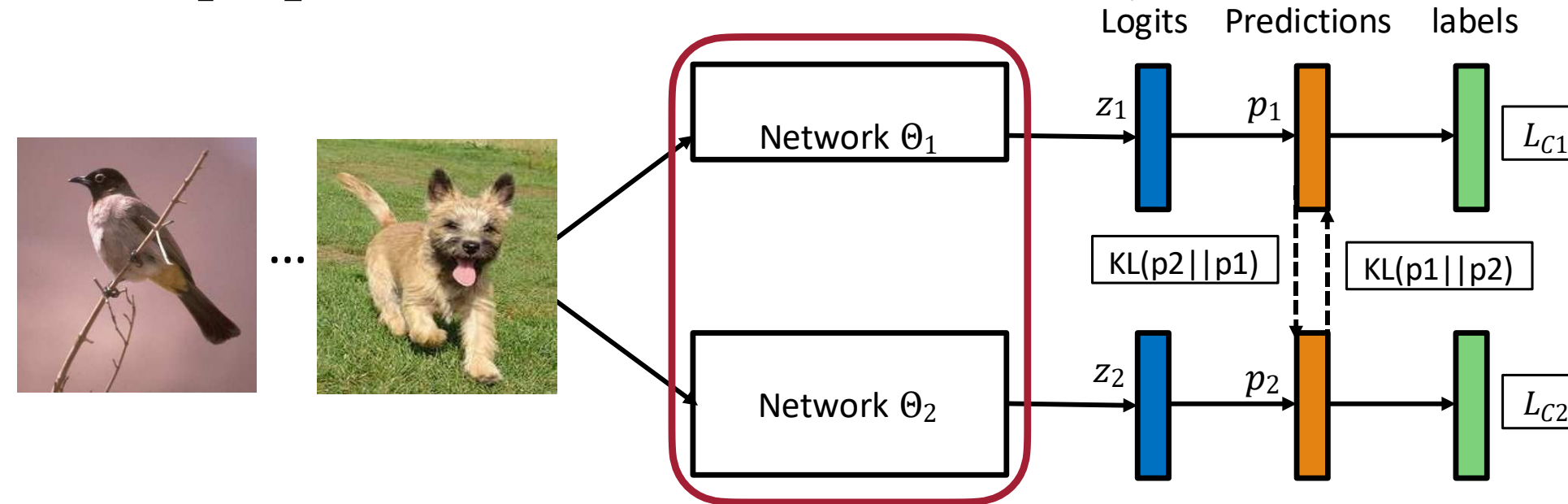
Online Distillation

Deep Mutual Learning

hier trainen 2 netwerken tegelijk
elk netw leert van:

- ground thruth labels
- output distribution van het andere netwerk

Θ_1, Θ_2 can be the same or different, and they are trained from scratch.



- Idea of deep mutual learning: for both teacher and student networks, we want to add a distillation objective that minimizes the output distribution of the other party.
- $\mathcal{L}(S) = \text{CrossEntropy}(S(I), y) + \text{KL}(S(I), T(I));$
- $\mathcal{L}(T) = \text{CrossEntropy}(T(I), y) + \text{KL}(T(I), S(I)).$
- Note: it is not necessary to pretrain T and $S = T$ is allowed.

Online Distillation

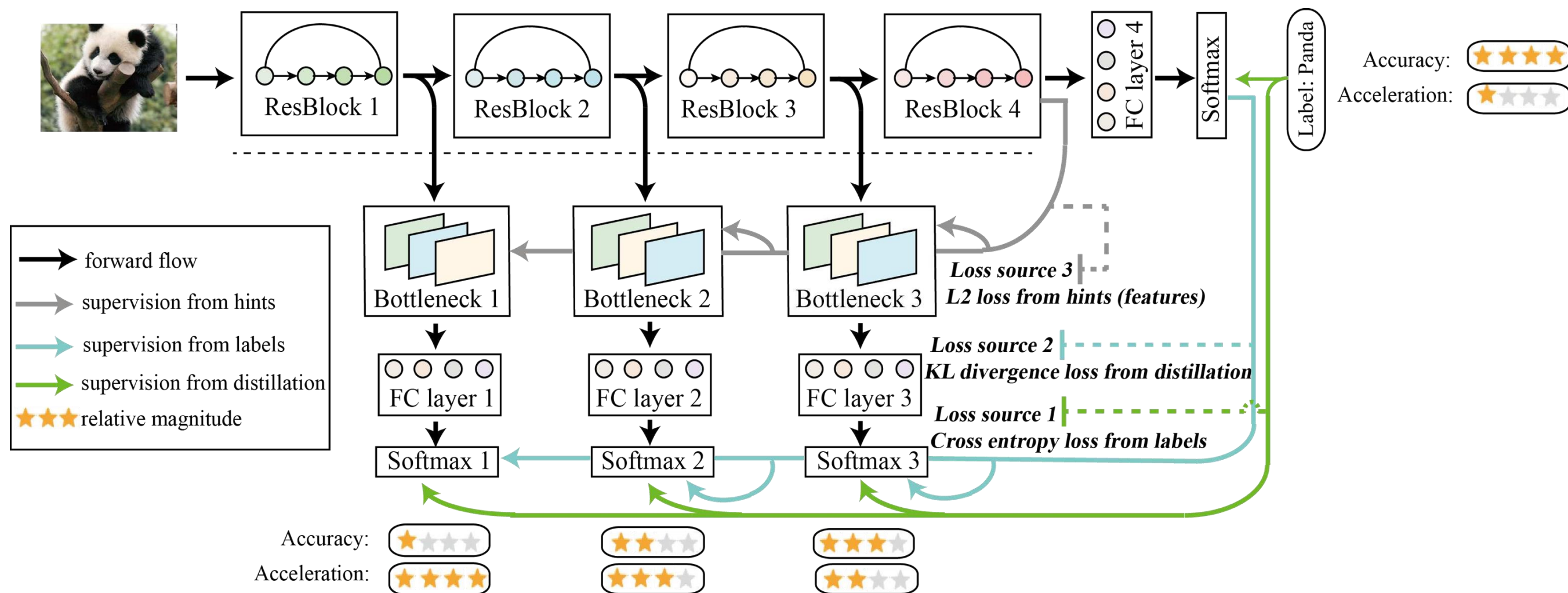
Deep Mutual Learning

Network Types		CIFAR-10						CIFAR-100					
		Independent		DML		DML-Ind		Independent		DML		DML-Ind	
Net 1	Net 2	Net 1	Net 2	Net 1	Net 2	Net 1	Net 2	Net 1	Net 2	Net 1	Net 2	Net 1	Net 2
Resnet-32	Resnet-32	92.47	92.47	92.68	92.80	0.21	0.33	68.99	68.99	71.19	70.75	2.20	1.76
WRN-28-10	Resnet-32	95.01	92.47	95.75	93.18	0.74	0.71	78.69	68.99	78.96	70.73	0.27	1.74
MobileNet	Resnet-32	93.59	92.47	94.24	93.32	0.65	0.85	73.65	68.99	76.13	71.10	2.48	2.11
MobileNet	MobileNet	93.59	93.59	94.10	94.30	0.51	0.71	73.65	73.65	76.21	76.10	2.56	2.45
WRN-28-10	MobileNet	95.01	93.59	95.73	94.37	0.72	0.78	78.69	73.65	80.28	77.39	1.59	3.74
WRN-28-10	WRN-28-10	95.01	95.01	95.66	95.63	0.65	0.62	78.69	78.69	80.28	80.08	1.59	1.39

Deep mutual learning can improve both student (net 2) and teacher (net 1) models.

Combining Online and Self-Distillation

Be Your Own Teacher: deep supervision + distillation



- Use deeper layers to distill shallower layers.
- **Intuition:** Labels at later stages are more reliable, so the authors use them to supervise the predictions from the previous stages.

Combining Online and Self-Distillation

Be Your Own Teacher: deep supervision + distillation

Neural Networks	Baseline	Classifier 1/4	Classifier 2/4	Classifier3/4	Classifier 4/4	Ensemble
VGG19(BN)	64.47	63.59	67.04	68.03	67.73	68.54
ResNet18	77.09	67.85	74.57	78.23	78.64	79.67
ResNet50	77.68	68.23	74.21	75.23	80.56	81.04
ResNet101	77.98	69.45	77.29	81.17	81.23	82.03
ResNet152	79.21	68.84	78.72	81.43	81.61	82.29
ResNeXt29-8	81.29	71.15	79.00	81.48	81.51	81.90
WideResNet20-8	79.76	68.85	78.15	80.98	80.92	81.38
WideResNet44-8	79.93	72.54	81.15	81.96	82.09	82.61
WideResNet28-12	80.07	71.21	80.86	81.58	81.59	82.09
PyramidNet101-240	81.12	69.23	78.15	80.98	82.30	83.51

- Results on CIFAR100 shows consistent performance improvements over the baseline.
- In addition, predictions from intermediate classifiers (1/4, 2/4, 3/4) can sometimes outperform the baseline. As such, inference efficiency can be improved.

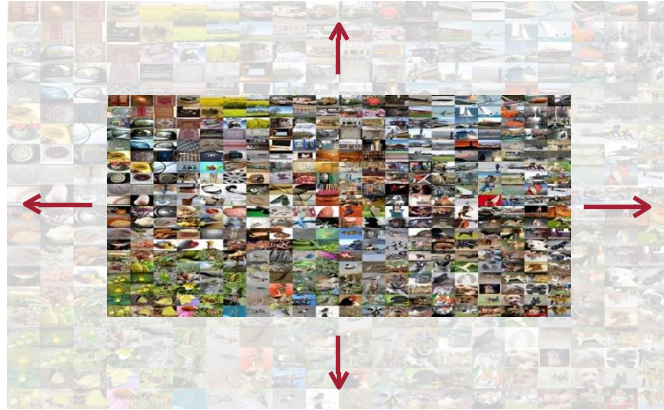
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- **Network augmentation**

Conventional Approach

Data augmentation during training to avoid overfitting

Data
Augmentation



Conventional Approach

Data augmentation during training to avoid overfitting

Data
Augmentation



Cutout



Conventional Approach

Data augmentation during training to avoid overfitting

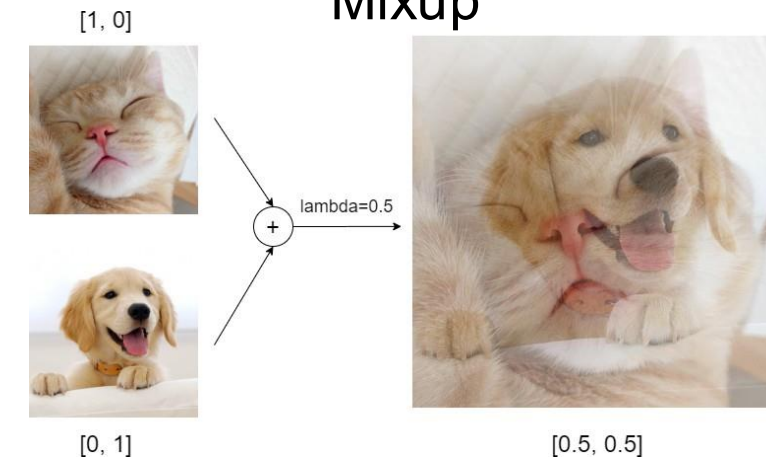
Data
Augmentation



Cutout



Mixup



Conventional Approach

Data augmentation during training to avoid overfitting

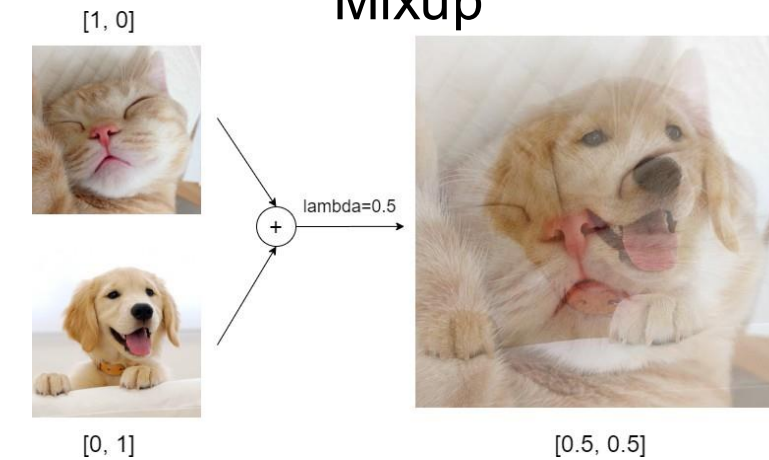
Data
Augmentation



Cutout



Mixup



AutoAugment



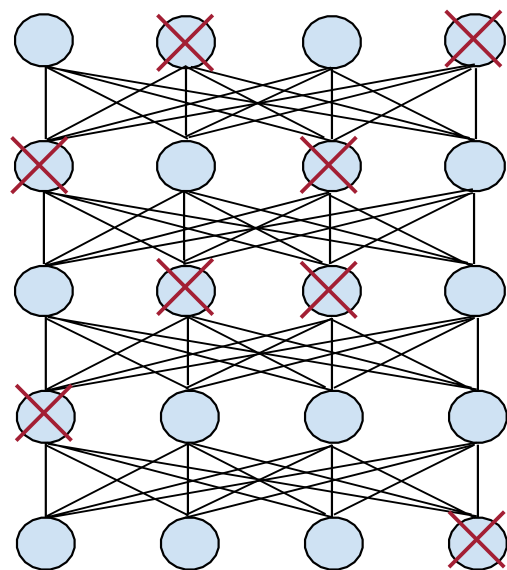
Conventional Approach

Dropout during training to avoid overfitting

Data
Augmentation



Dropout



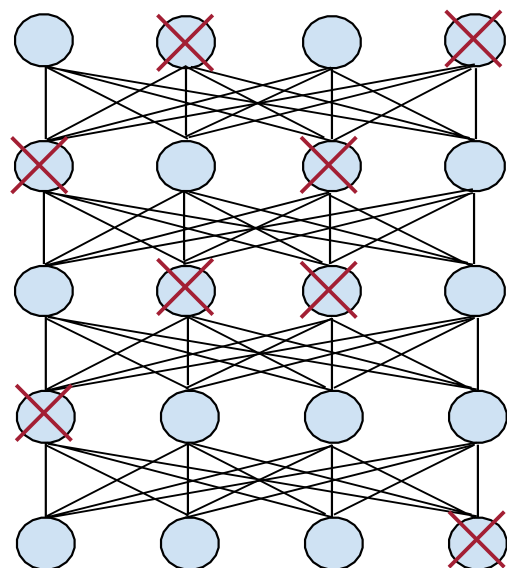
Conventional Approach

Dropout during training to avoid overfitting

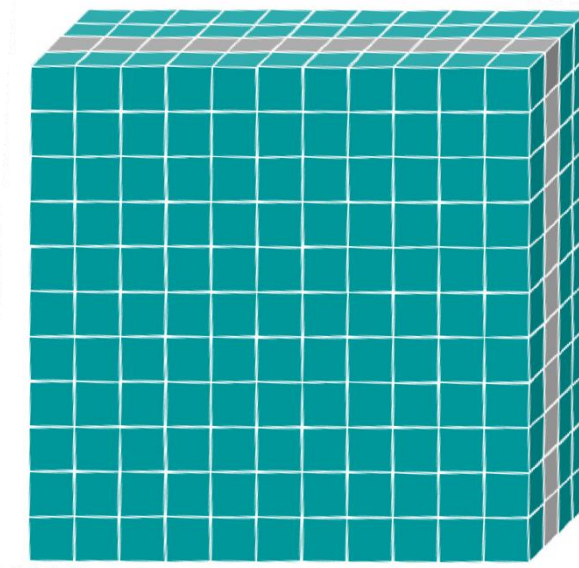
Data
Augmentation



Dropout



SpatialDropout



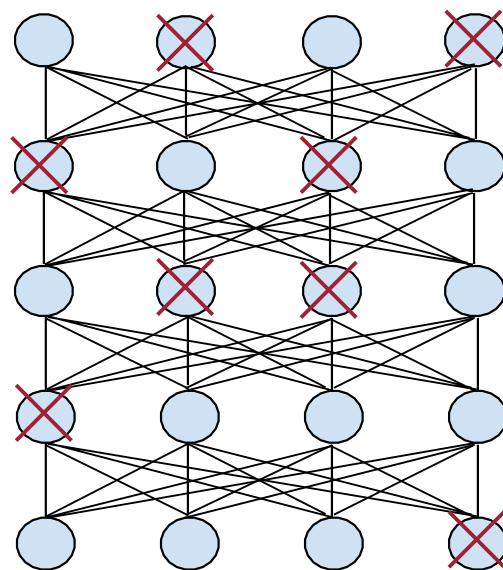
Conventional Approach

Dropout during training to avoid overfitting

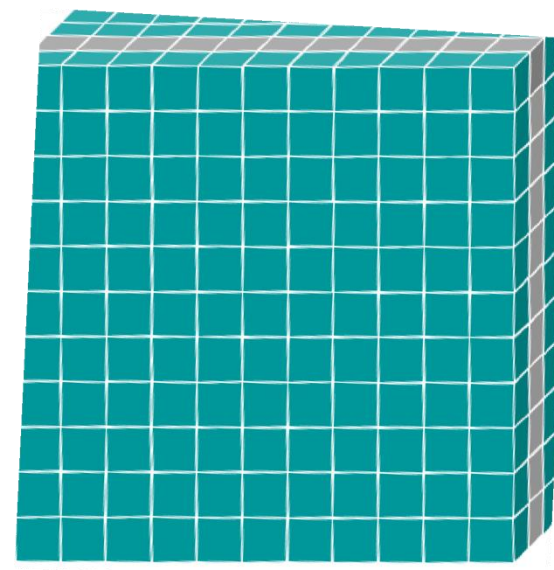
Data
Augmentation



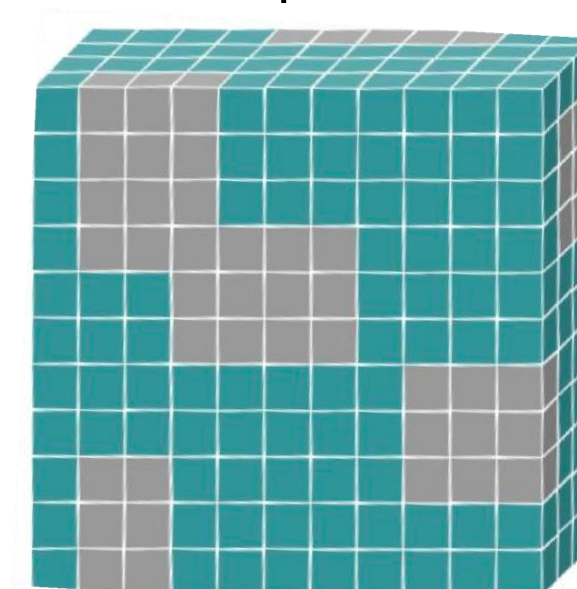
Dropout



SpatialDropout



DropBlock



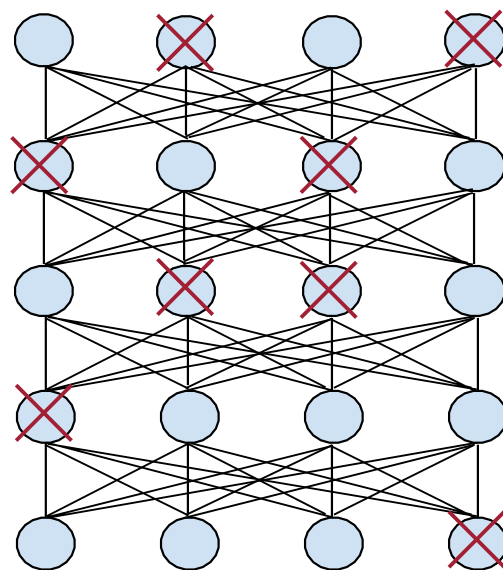
Conventional Approach

Dropout/Data augmentation improves large neural networks' performance

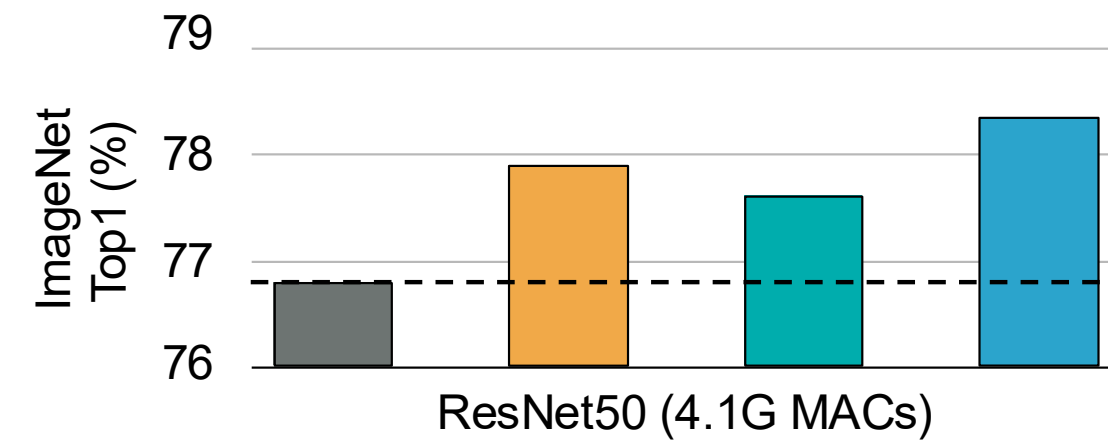
Data
Augmentation



Dropout



Baseline Mixup AutoAugment DropBlock



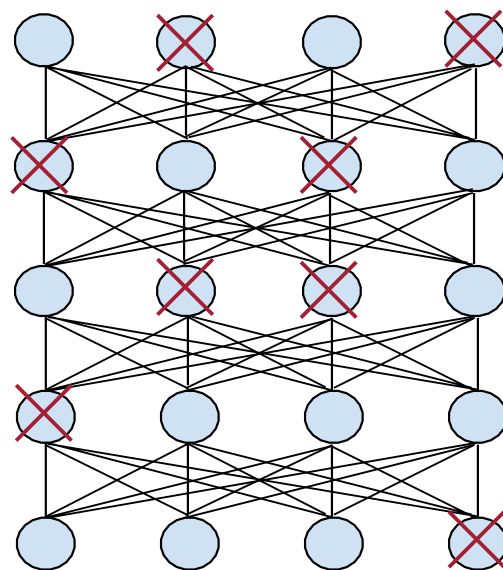
Conventional Approach

Dropout/Data augmentation hurts tiny neural networks' performance

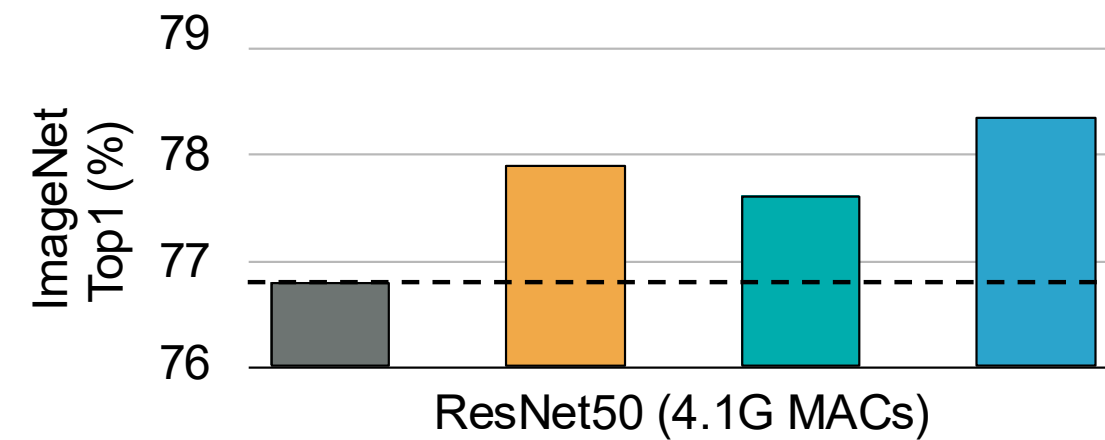
Data
Augmentation



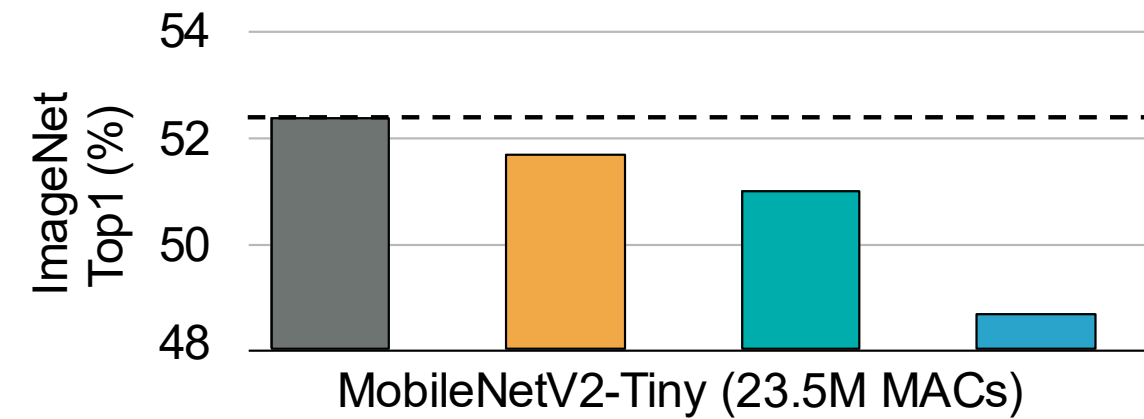
Dropout



Baseline Mixup AutoAugment DropBlock



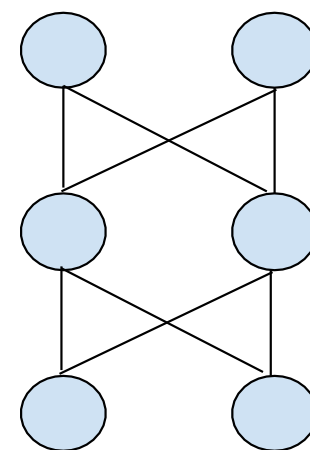
Baseline Mixup AutoAugment Dropblock



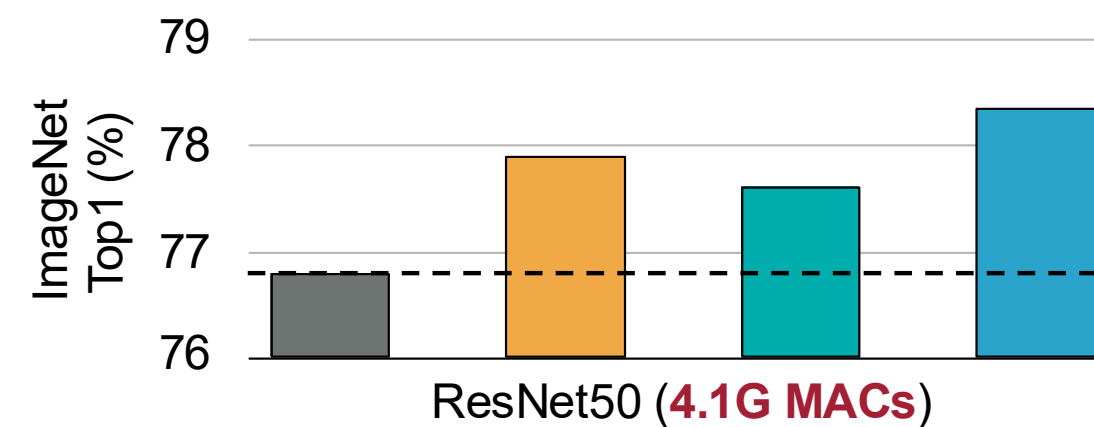
Conventional Approach

Tiny neural network lacks capacity

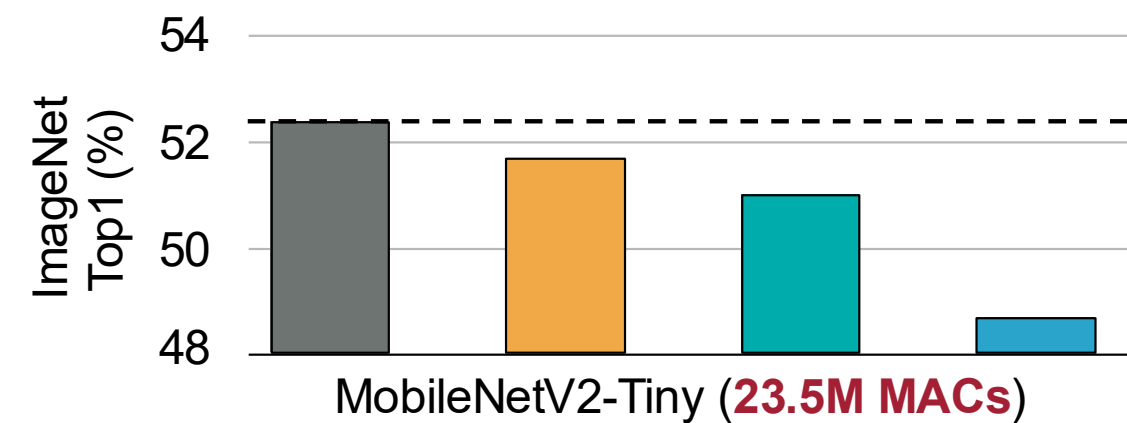
Tiny Neural Network



Baseline Mixup AutoAugment DropBlock

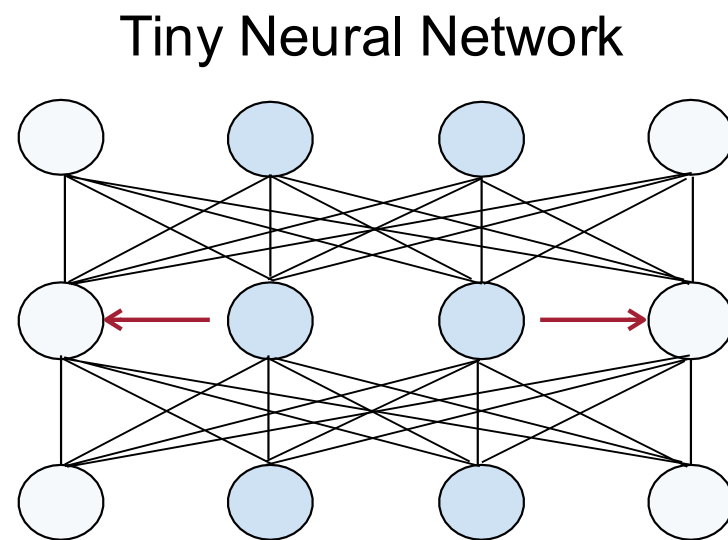


Baseline Mixup AutoAugment Dropblock

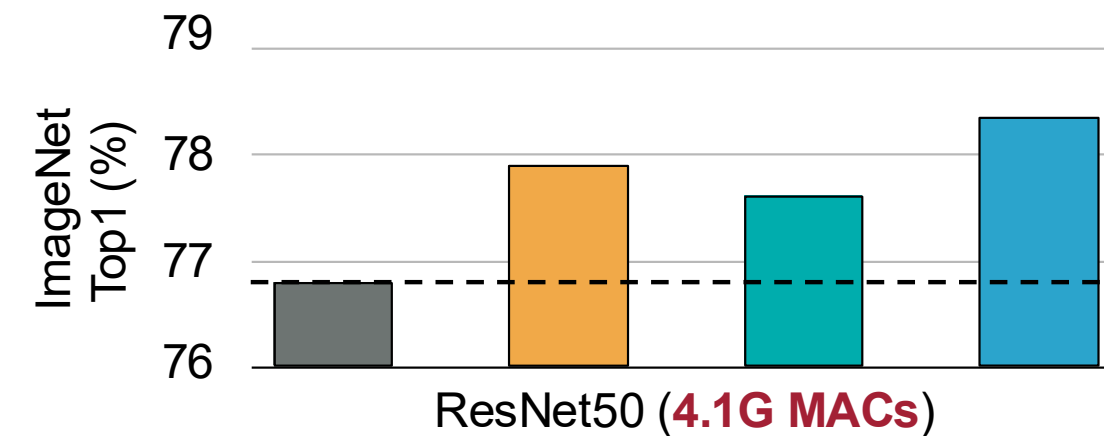


Network Augmentation

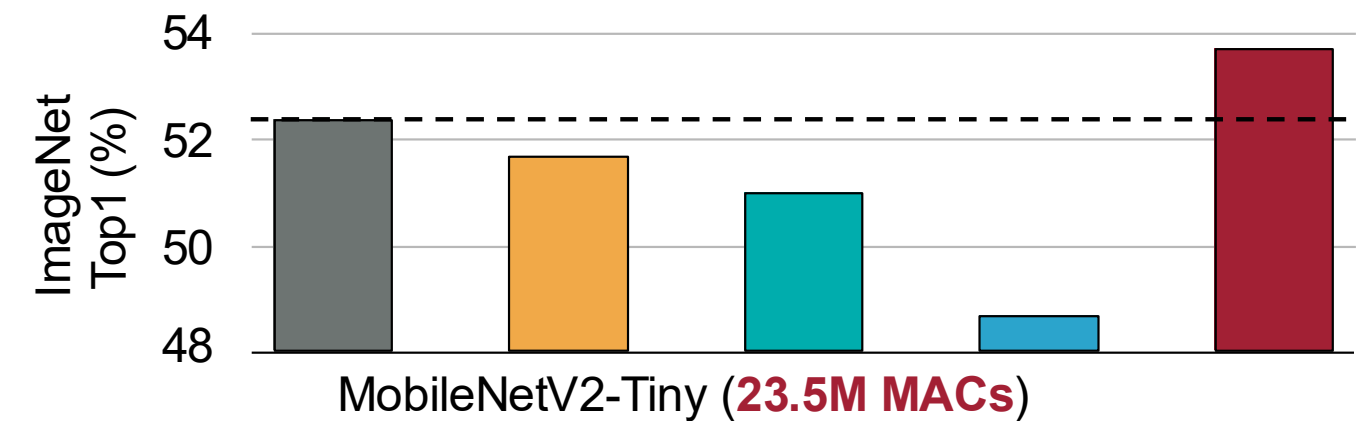
Augment the model to get extra supervision during training for tiny models



Baseline Mixup AutoAugment DropBlock

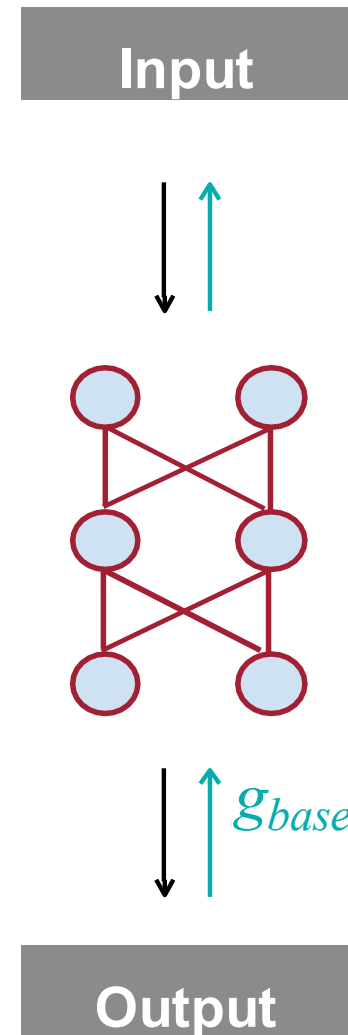


Baseline Mixup AutoAugment Dropblock NetAug



NetAug: Training Process

Augment the model to get extra supervision during training for tiny models

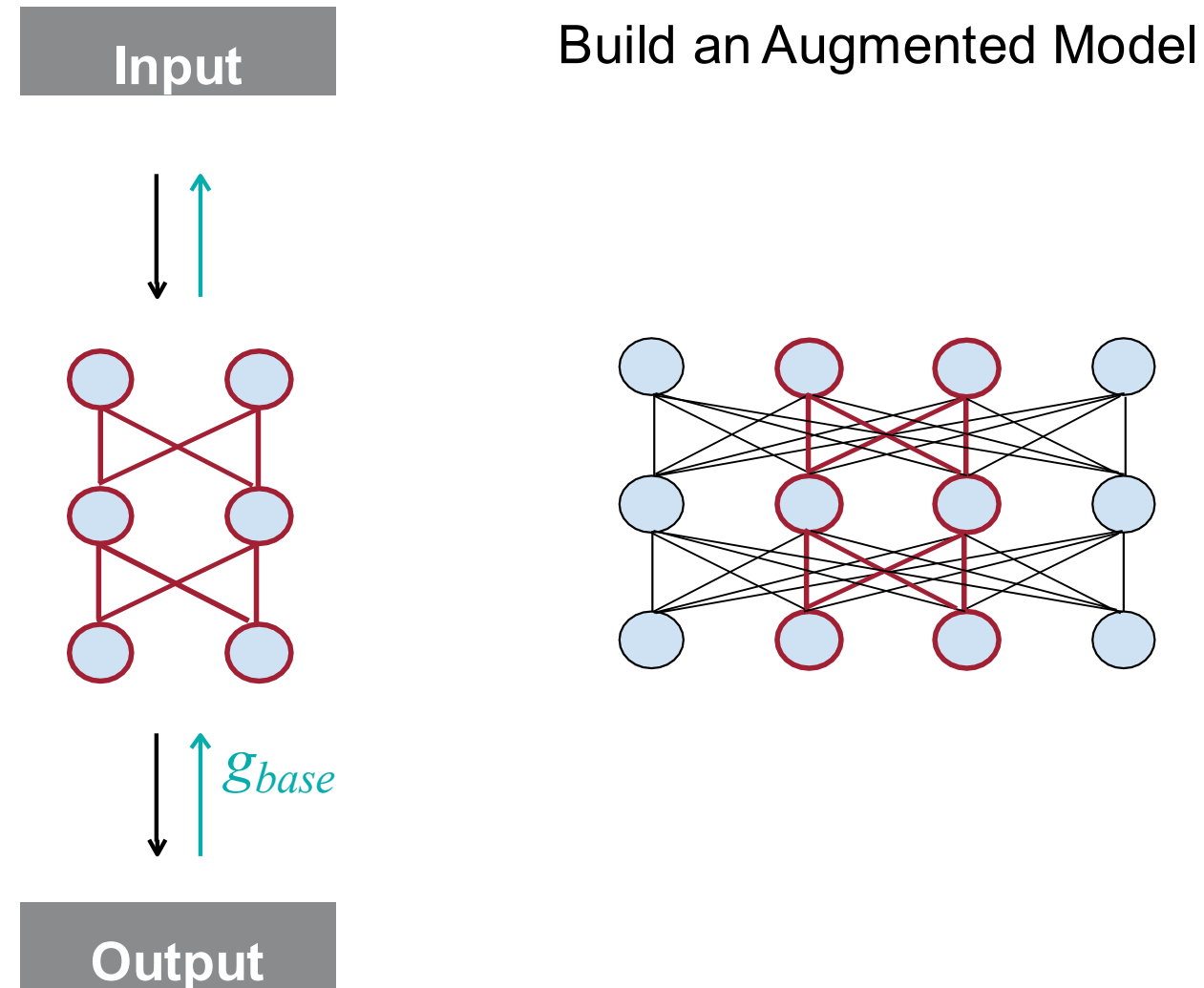


$$L_{aug} = \underbrace{L(W_{base})}_{\text{base supervision}}$$

Network Augmentation for Tiny Deep Learning [Cai *et al.*, ICLR 2022]

NetAug: Training Process

Augment the model to get extra supervision during training for tiny models

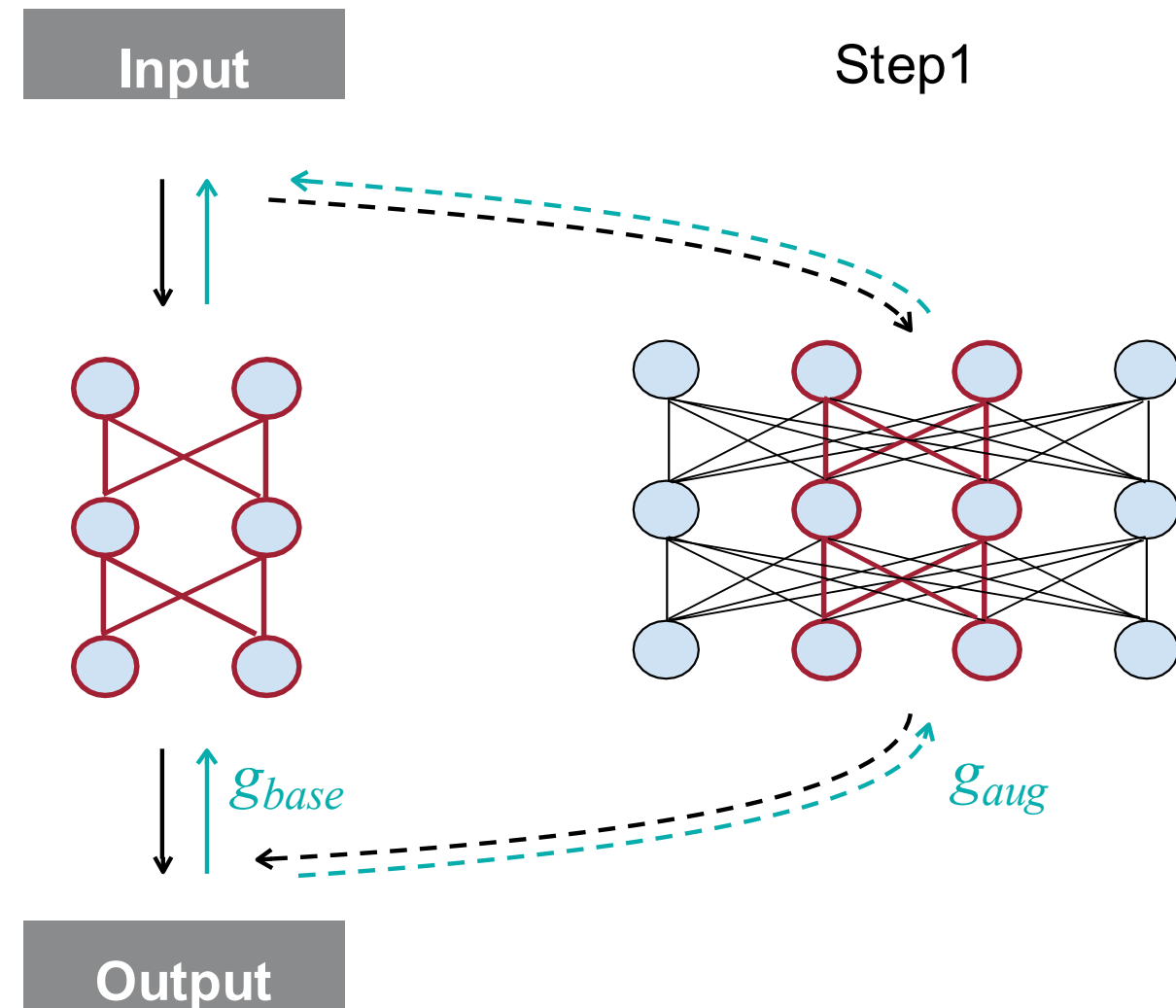


$$L_{\text{aug}} = \underbrace{L(W_{\text{base}})}_{\text{base supervision}}$$

Network Augmentation for Tiny Deep Learning [Cai *et al.*, ICLR 2022]

NetAug: Training Process

Augment the model to get extra supervision during training for tiny models

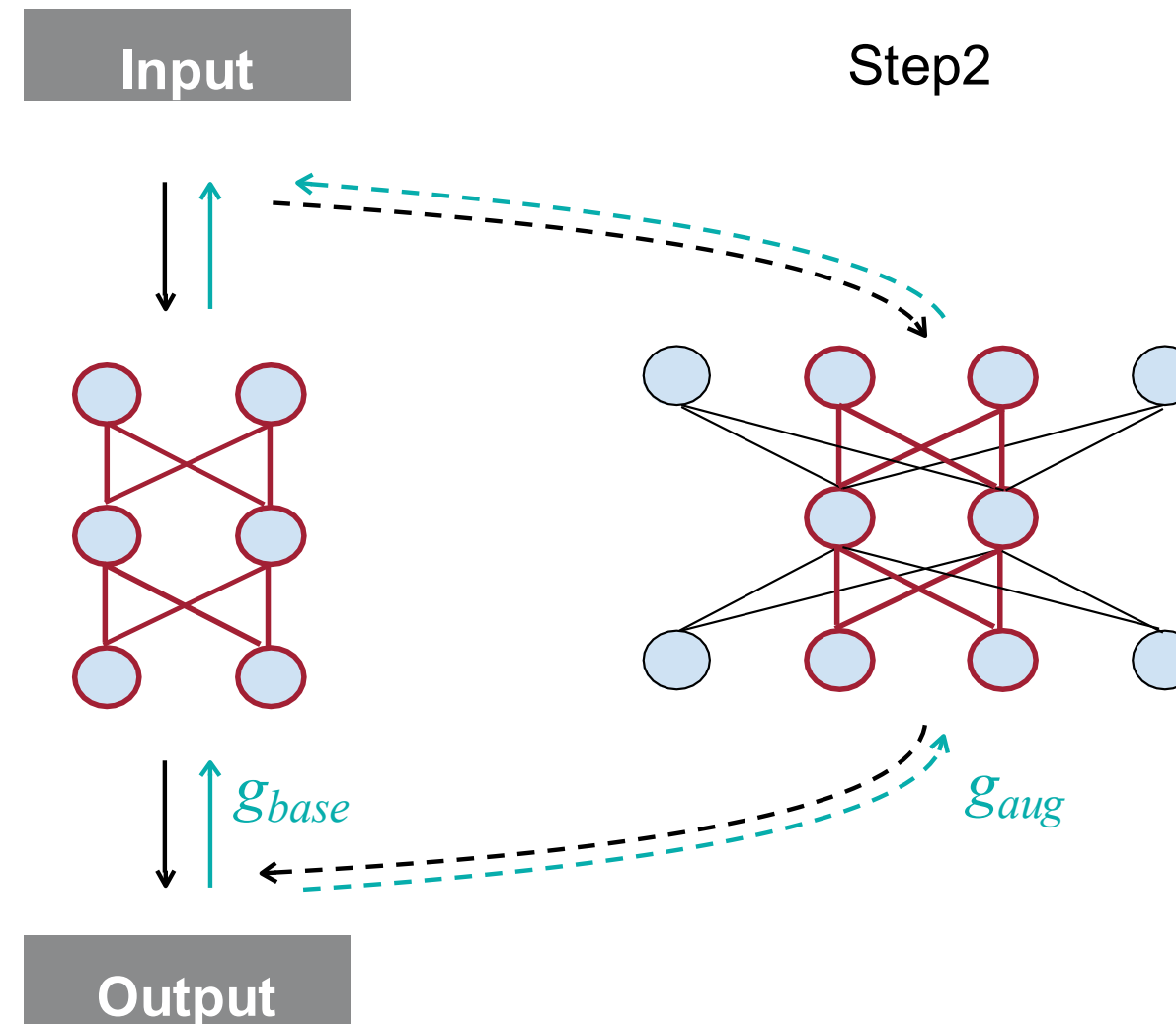


$$L_{aug} = \underbrace{L(\underbrace{W_{base}}_{\text{base supervision}})}_{\text{base supervision}} + \underbrace{\alpha L(\underbrace{[W_{base}, W_{aug}]}_{\text{auxiliary supervision}})}_{\text{auxiliary supervision}}$$

Network Augmentation for Tiny Deep Learning [Cai *et al.*, ICLR 2022]

NetAug: Training Process

Augment the model to get extra supervision during training for tiny models

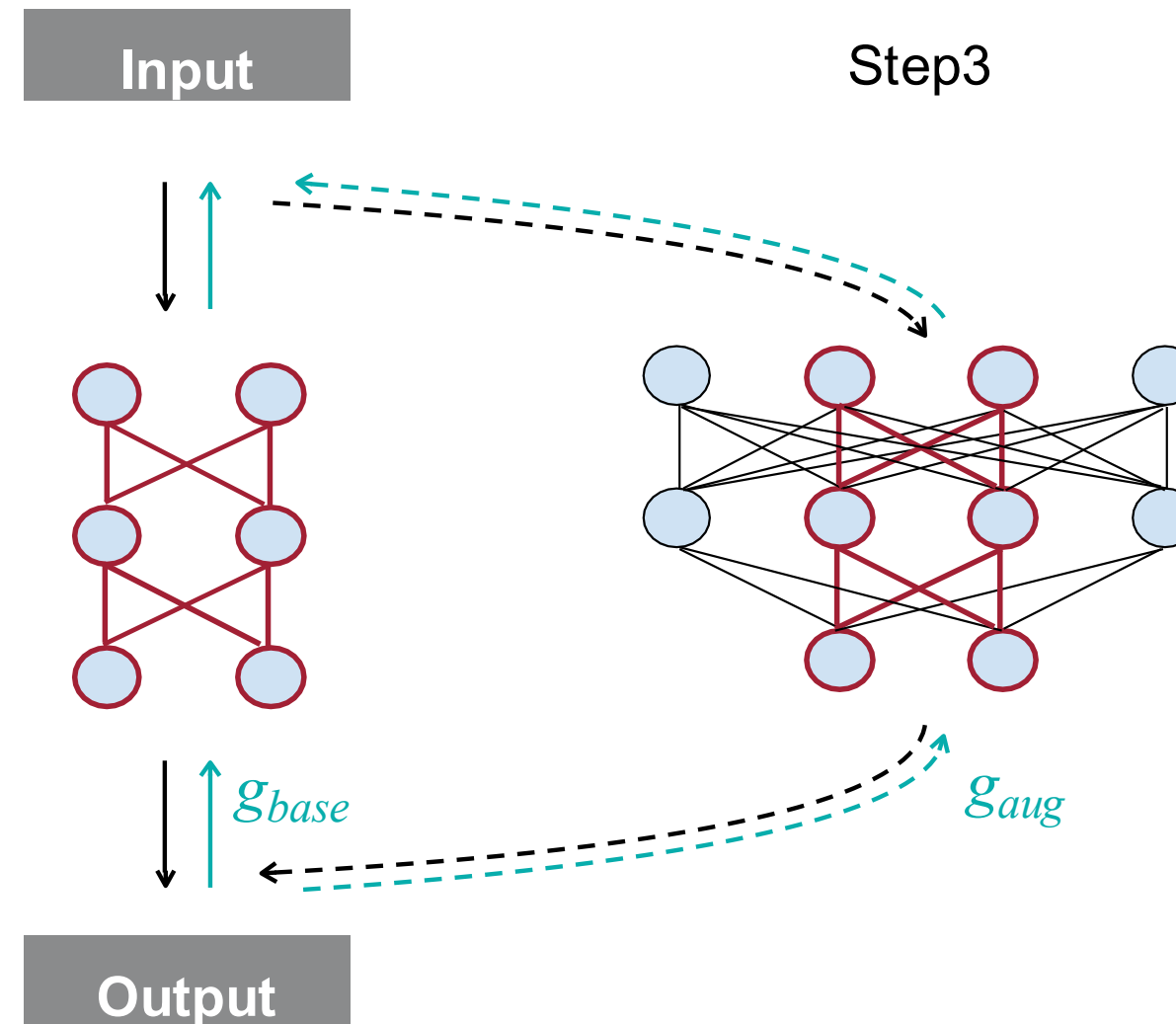


$$L_{\text{aug}} = \underbrace{L(W_{\text{base}})}_{\text{base supervision}} + \alpha \underbrace{L([W_{\text{base}}, W_{\text{aug}}])}_{\text{auxiliary supervision}}$$

Network Augmentation for Tiny Deep Learning [Cai *et al.*, ICLR 2022]

NetAug: Training Process

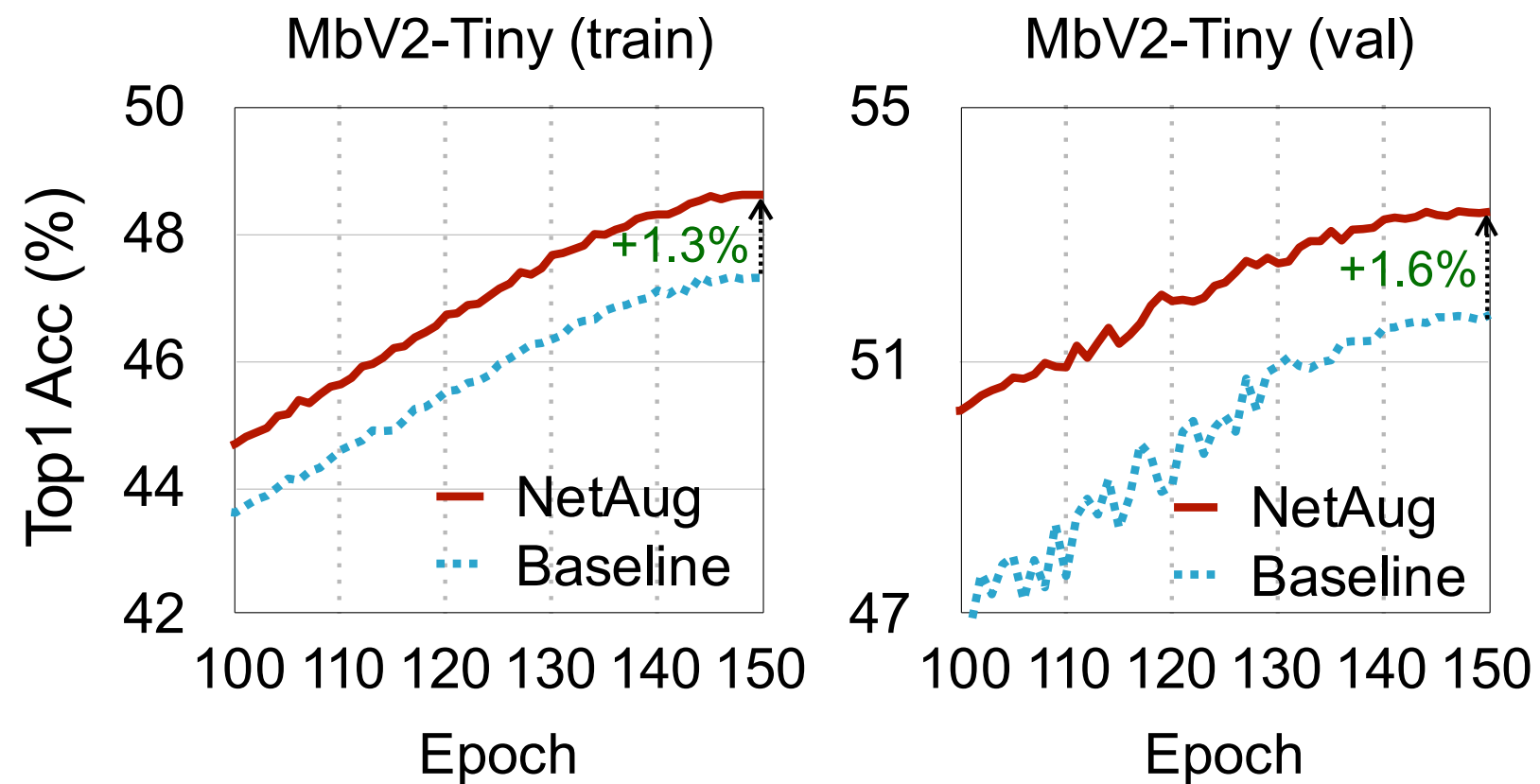
Augment the model to get extra supervision during training for tiny models



$$L_{aug} = \underbrace{L(\underbrace{W_{base}}_{\text{base supervision}})}_{\text{base supervision}} + \underbrace{\alpha L(\underbrace{[W_{base}, W_{aug}]}_{\text{auxiliary supervision}})}_{\text{auxiliary supervision}}$$

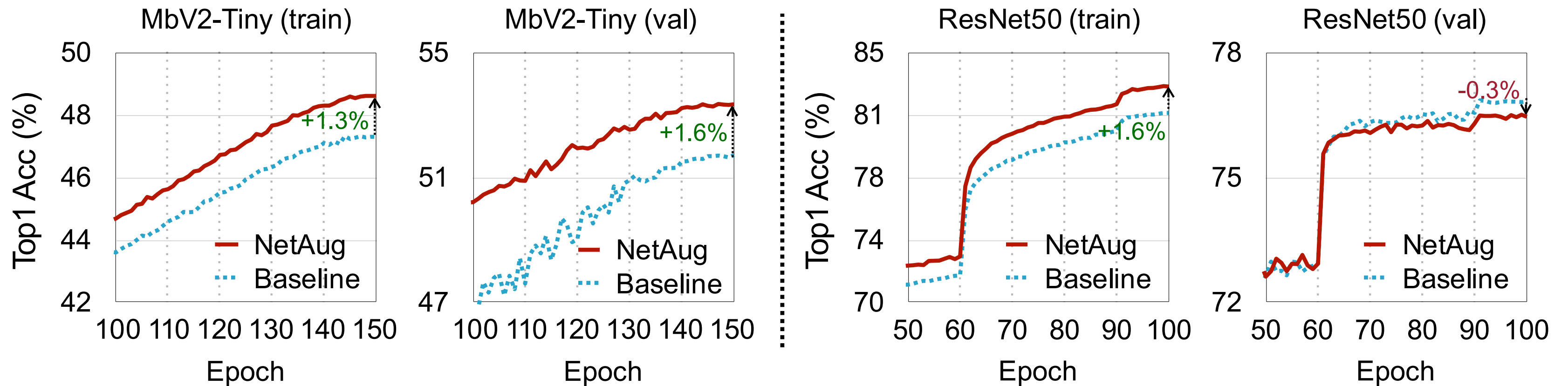
Network Augmentation for Tiny Deep Learning [Cai *et al.*, ICLR 2022]

NetAug: Learning Curve



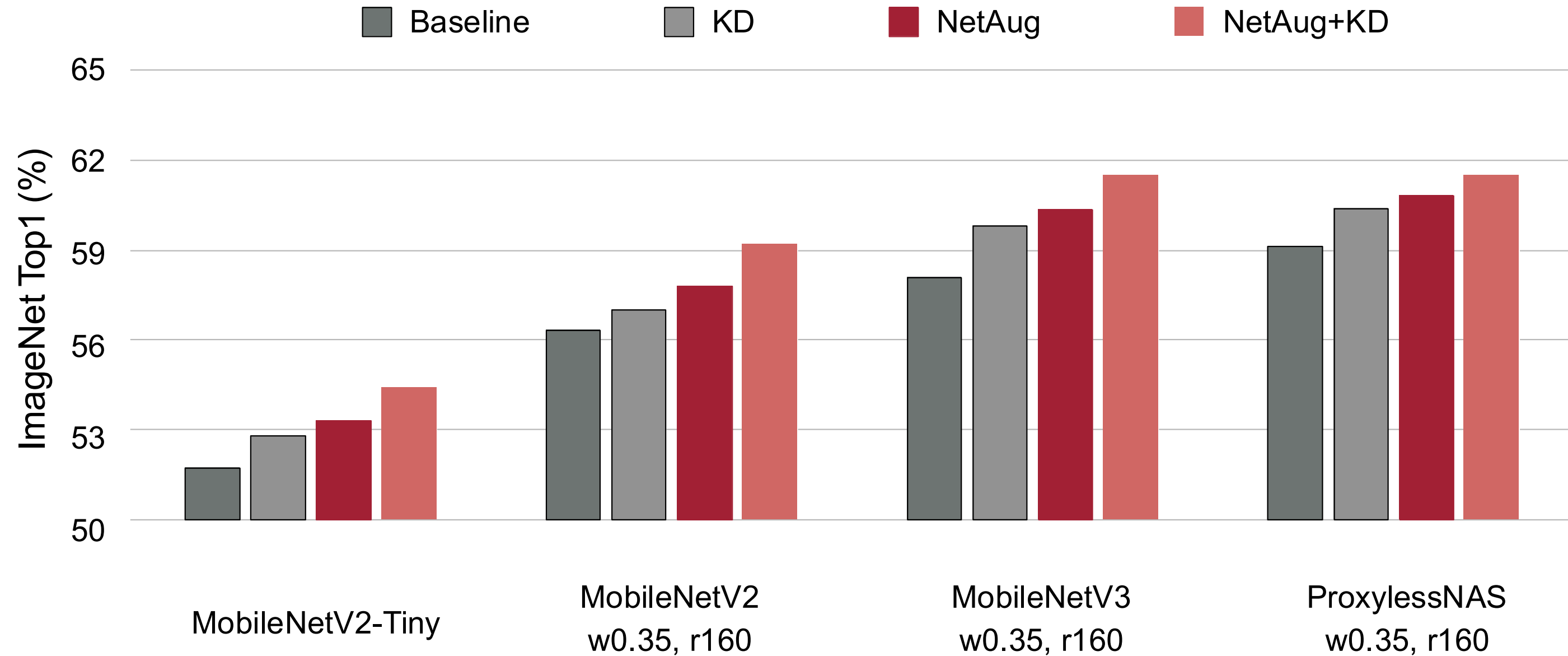
- For a tiny neural network, NetAug improves both the training accuracy and val accuracy.

NetAug: Learning Curve



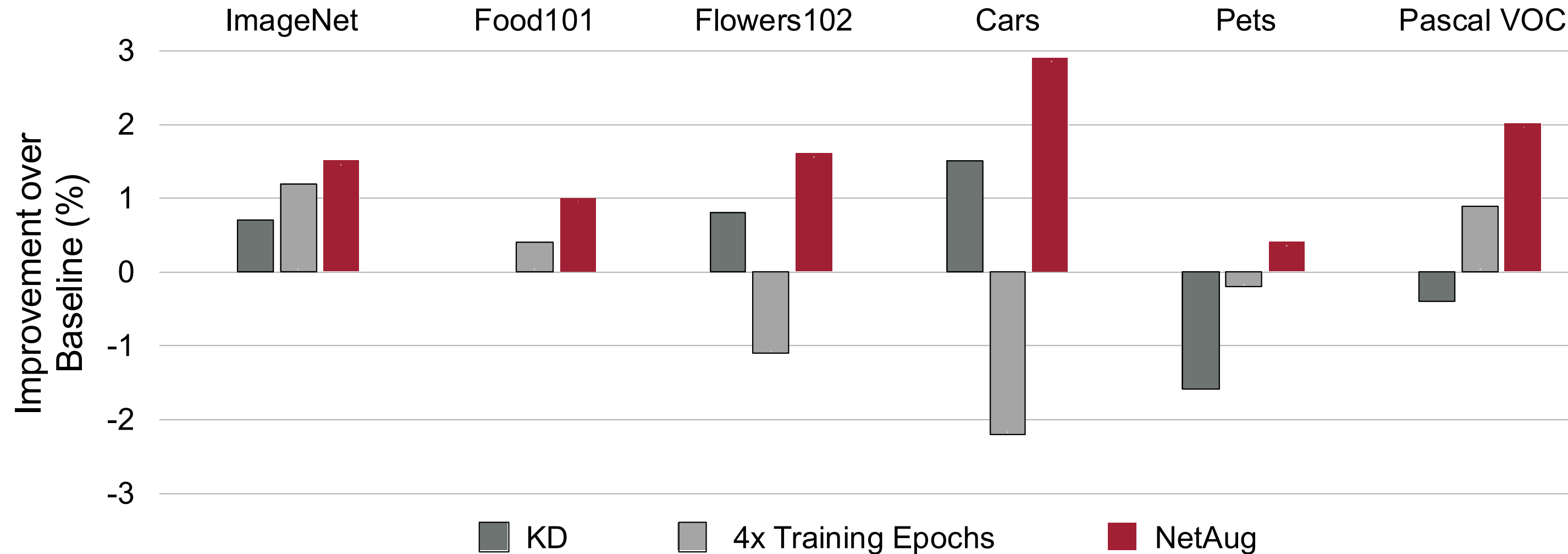
- For a tiny neural network, NetAug improves **both** the training accuracy and val accuracy.
- For a large neural network, NetAug **improves** the training accuracy but **hurts** the val accuracy.

NetAug: Results



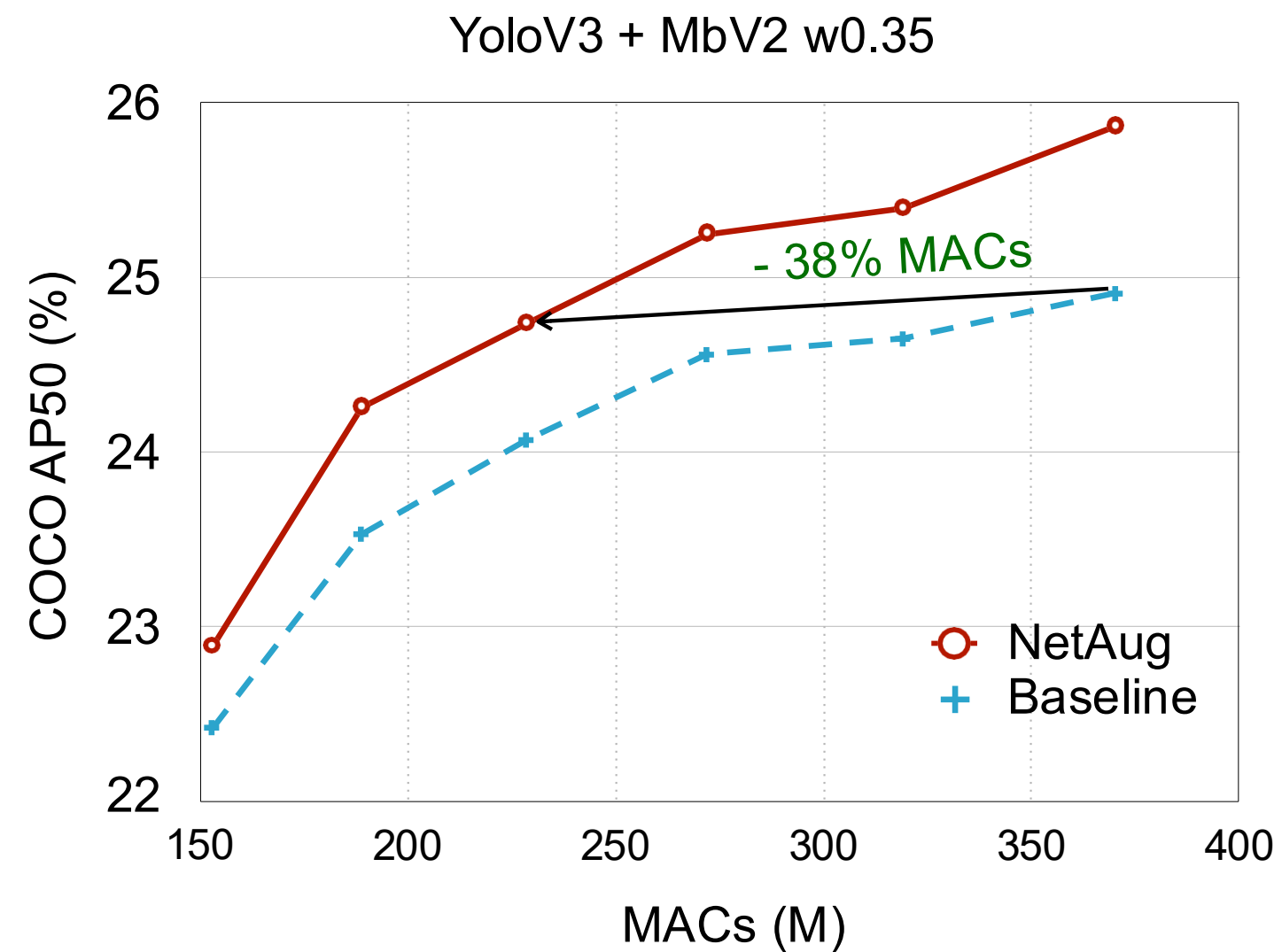
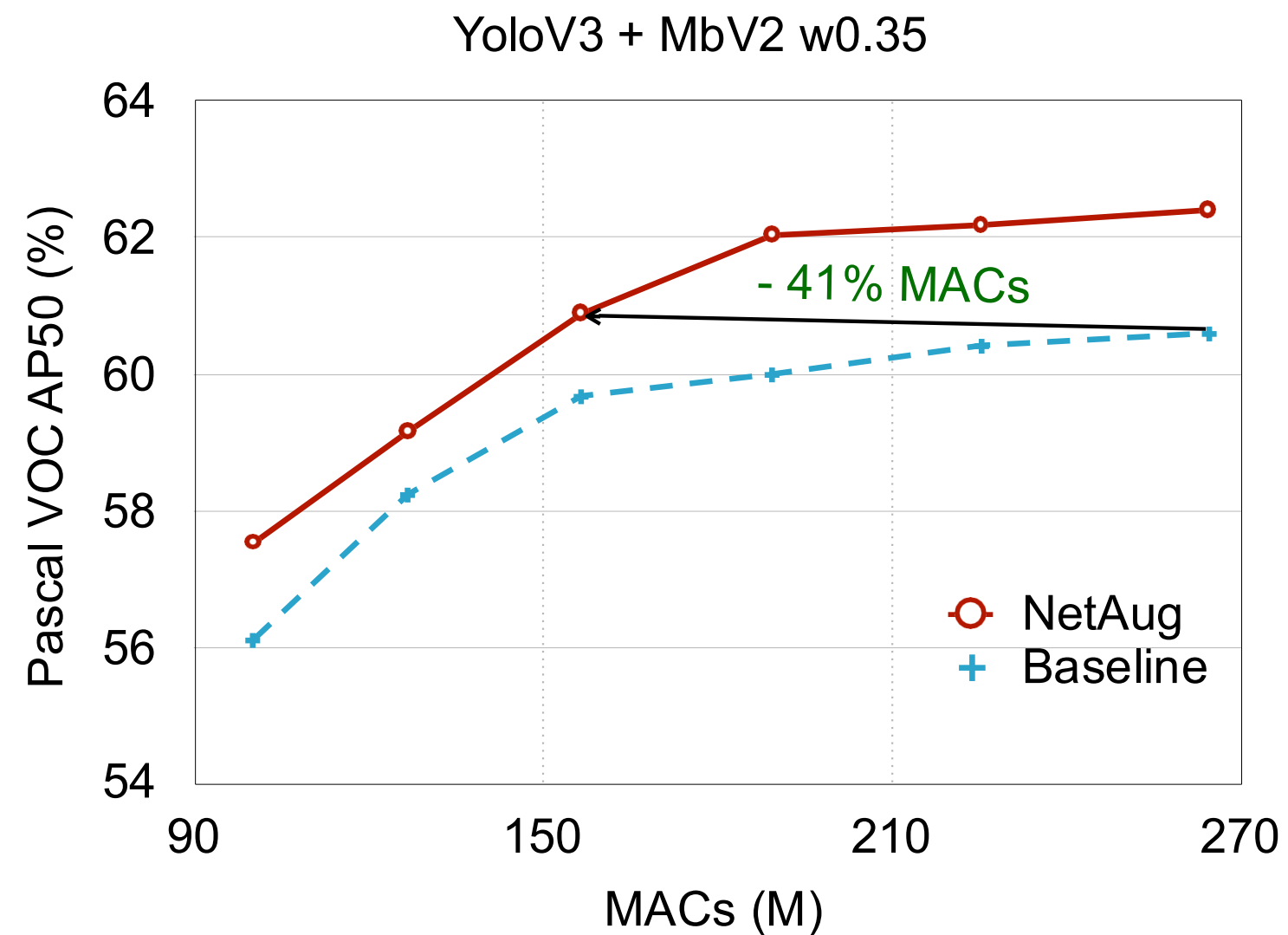
- NetAug is orthogonal to KD.

NetAug: Transfer Learning



- NetAug provides better transfer learning performances than KD and 4x training schedule, though their ImageNet performances are similar.

NetAug: Transfer to Object Detection



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